DFM_HG_V57_P1_FBW 12UP-Auto Machine

Version:V1.1

Wuhan HGlaser Engineering Co.,Ltd

HGTECH

Change list

Version	Notes	Page	Date
V1.0	First release		2024.07.12
V1.1	Remove and replace F75 collimation mirror		2024.07.24

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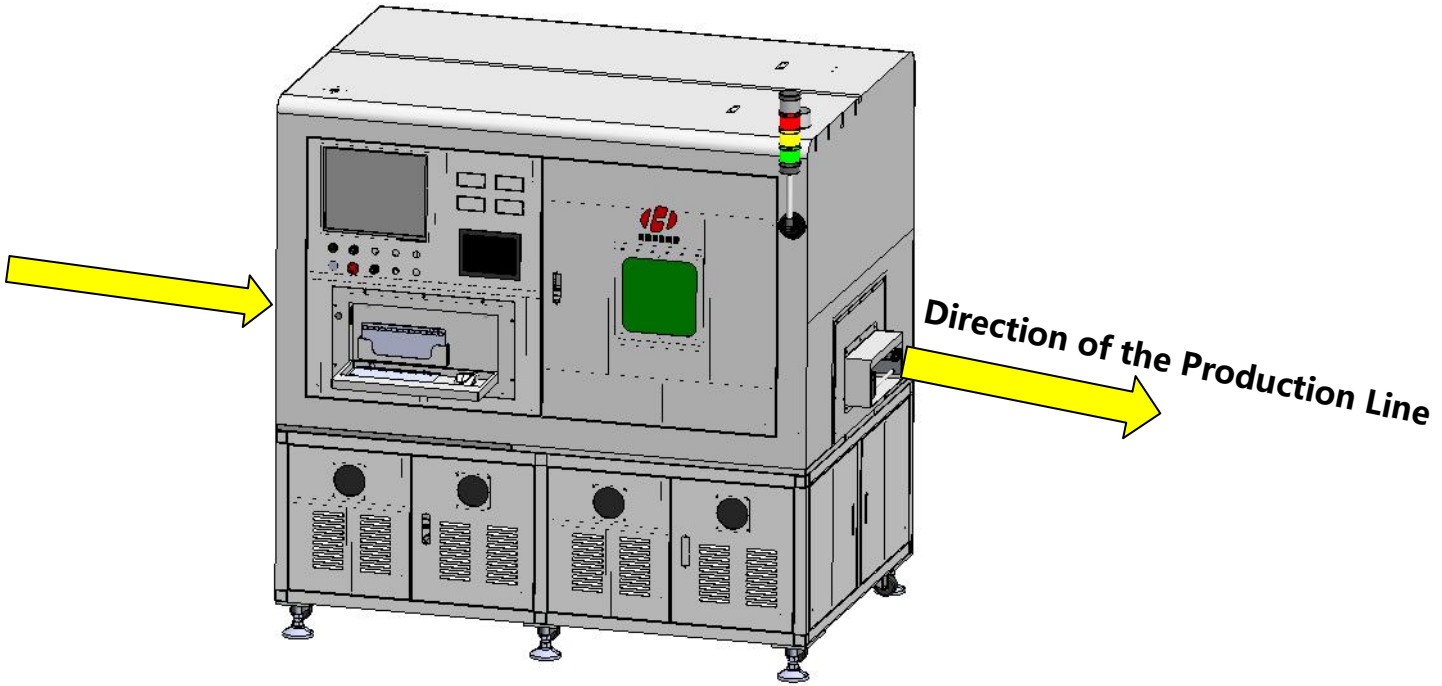
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9 Changes

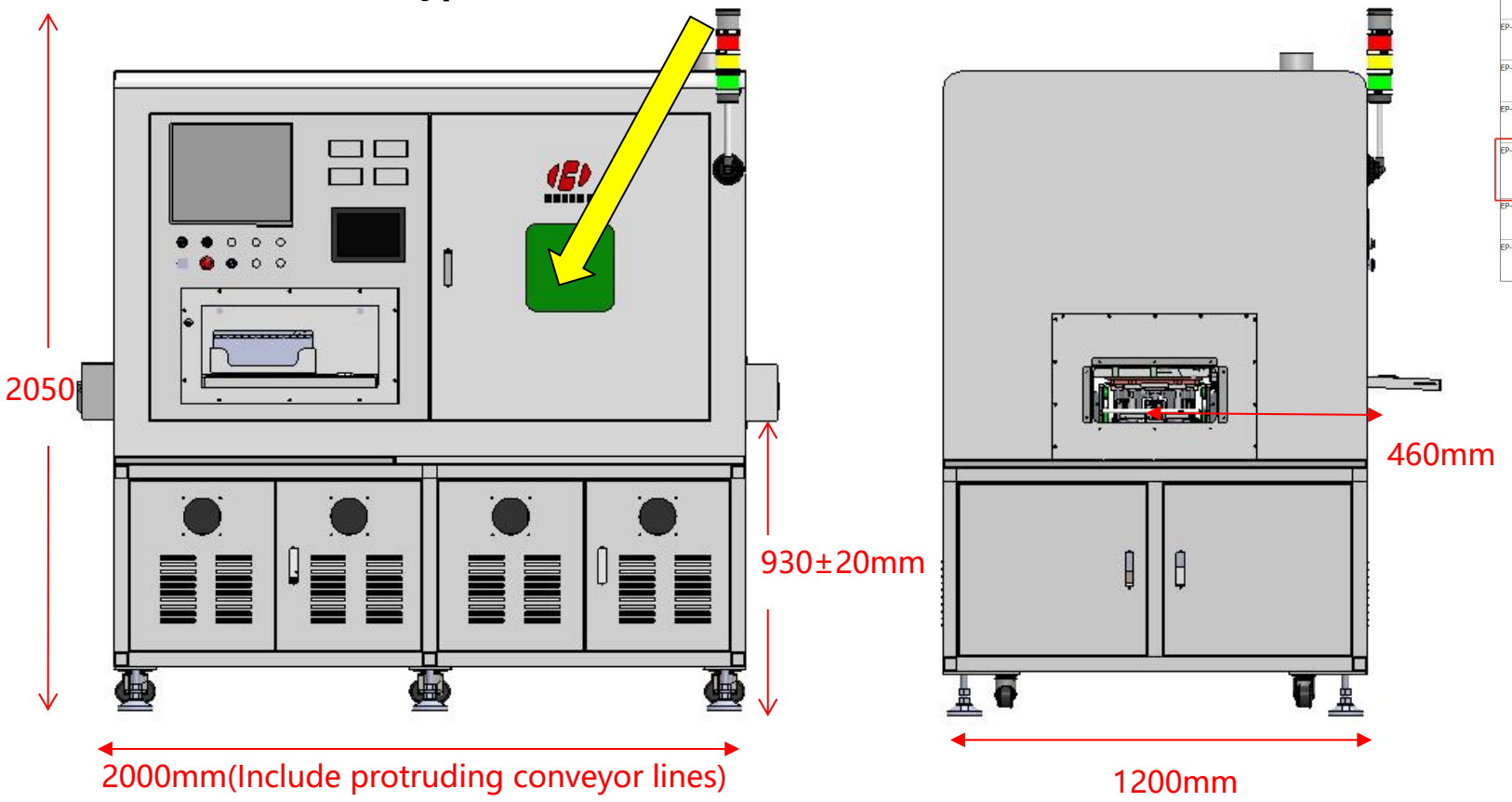
1 Overview of Machine



- Modular Design, the structure of system is clear, the commonality and expandability are strong.
- /LFW/LFW with the same design, one station is easy to change to another.
- One machine contains QR reading ,laser sensor and auto welding， auto welding contains double Galvanometer heads， 4 clamping cylinders.

2.1 Equipment Overview

Perspective window: 250*250mm
type EP5+

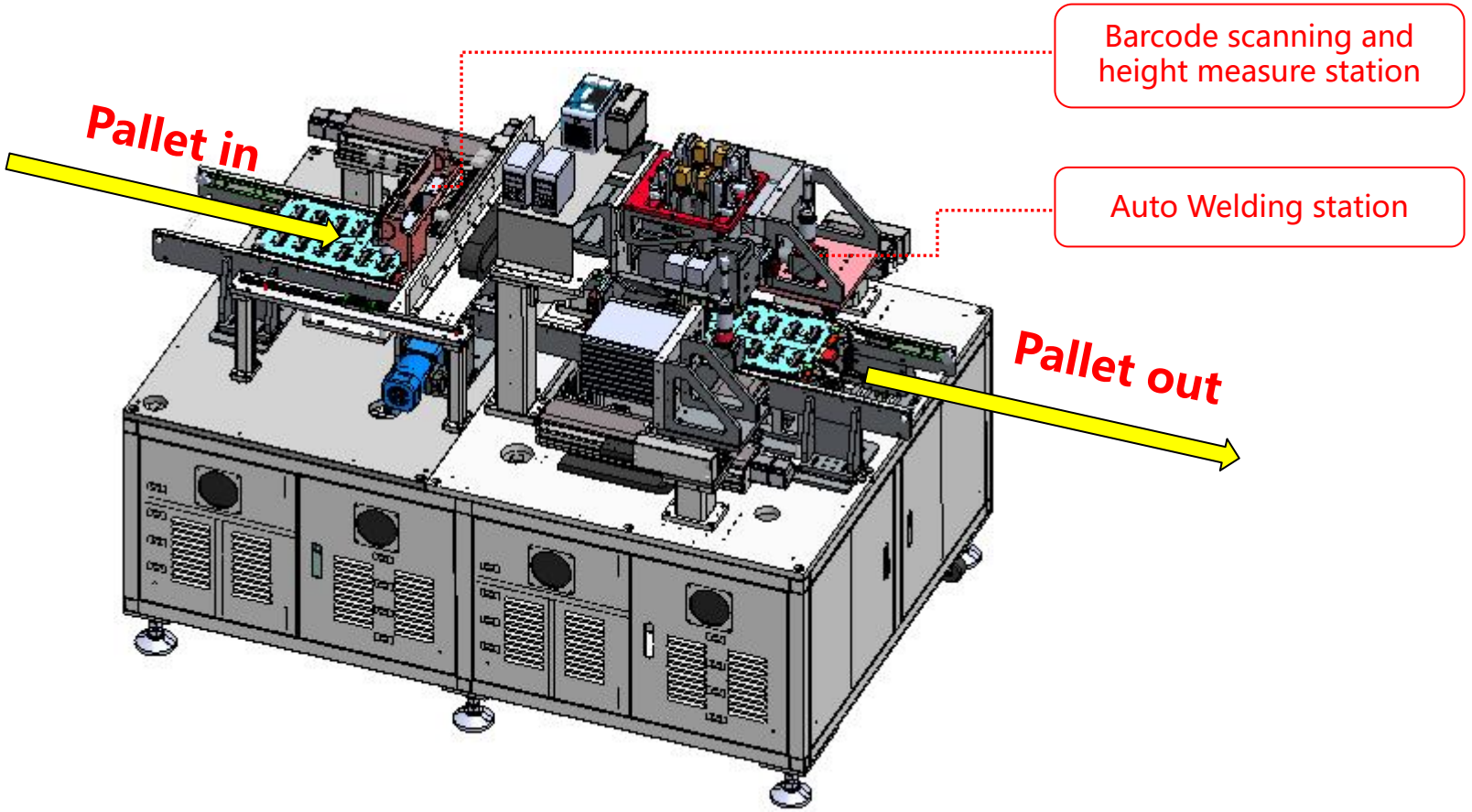


Eagle Pair Laser Safety Viewing Windows Specs:

Part Number	Size(mm)	Protected Wavelength & Optical Density
EP-1A	1000*1200*5	190-532&900-1700nm OD=4+
EP-2	1000*1200*5	632nm OD=4+
EP-3	1000*1200*5	190-532nm OD=4+
EP-4	1000*1200*5	10600nm OD=5+
EP-5	1000*1200*5	190-450&800-1700nm OD=4+
EP-8	1000*1200*5	750-1350nm OD=4+
EP-9	1000*1200*5	190-400nm OD=4+

RBW RFW LFW FBW

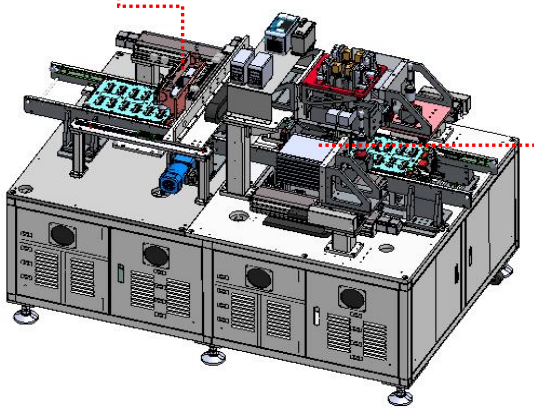
2.2 Kernel structure

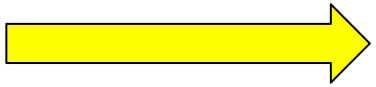


2.3 Work flow

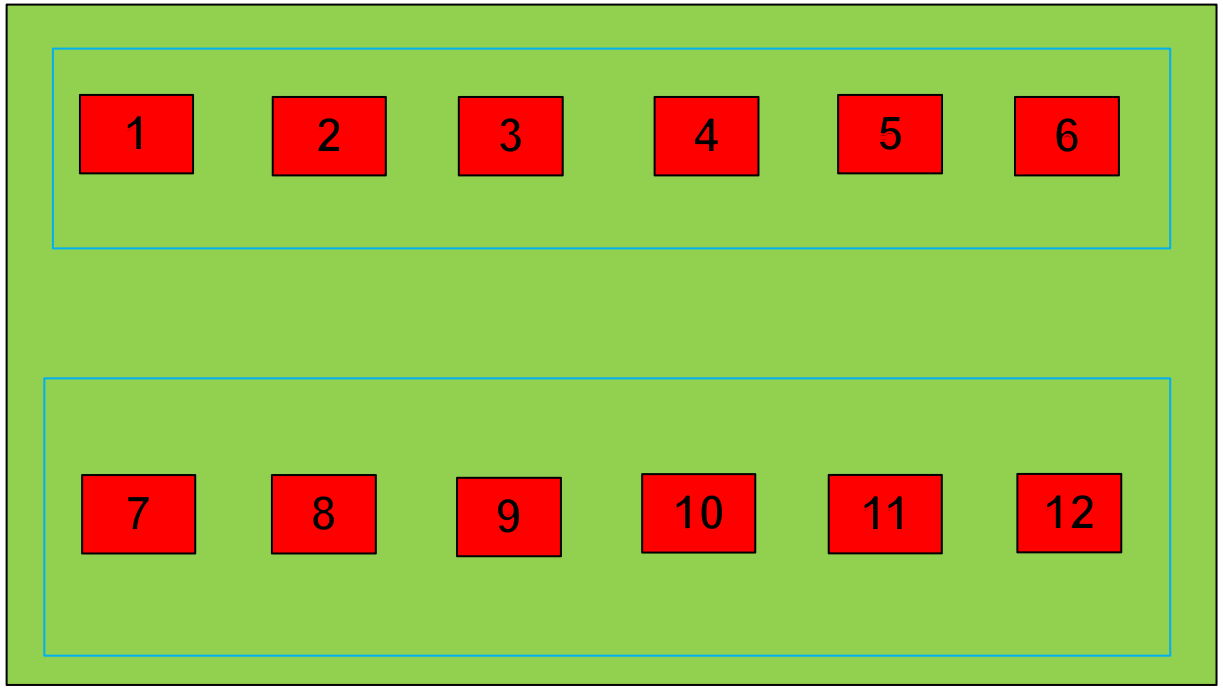
Barcode scanning and height measure station

Auto Welding station





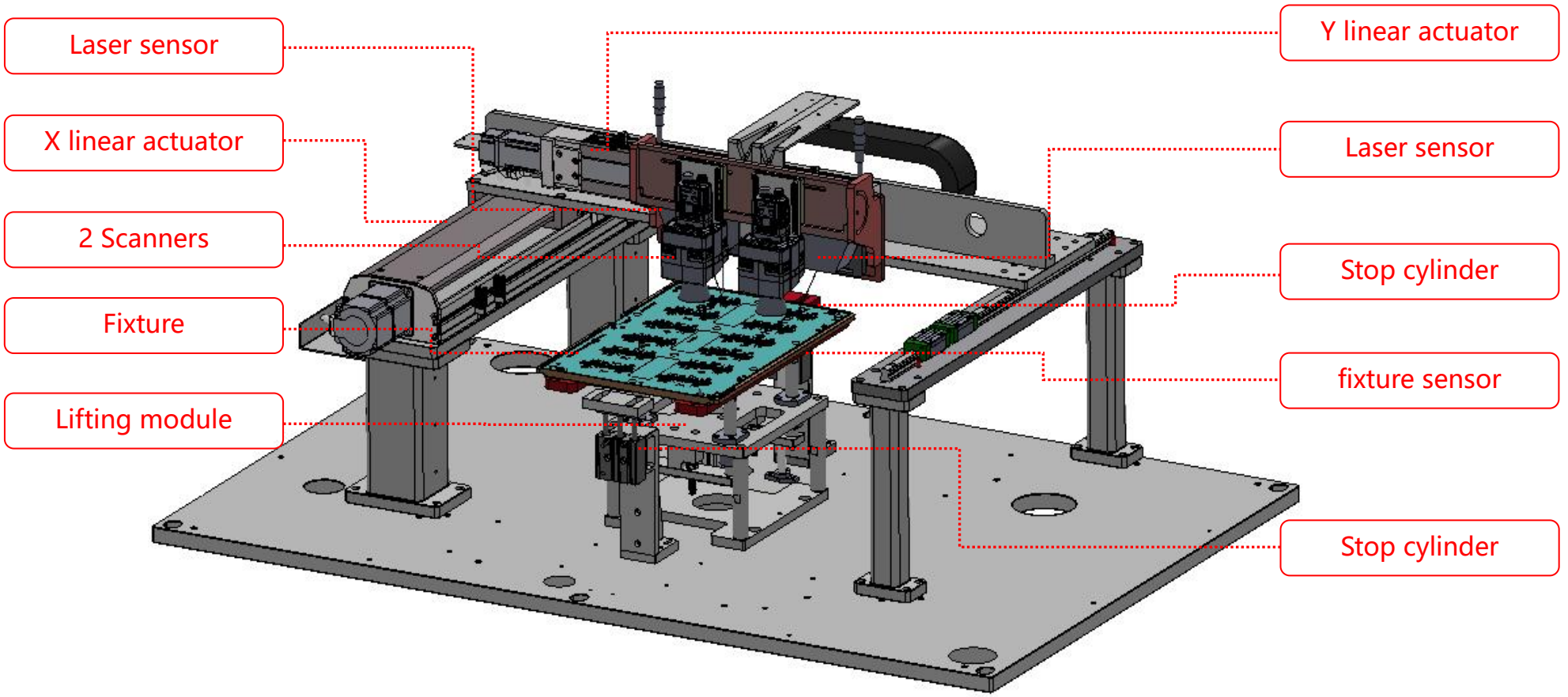
Galvanometer1



Galvanometer2

Galvanometer1 and Galvanometer2 work simultaneously to save time.

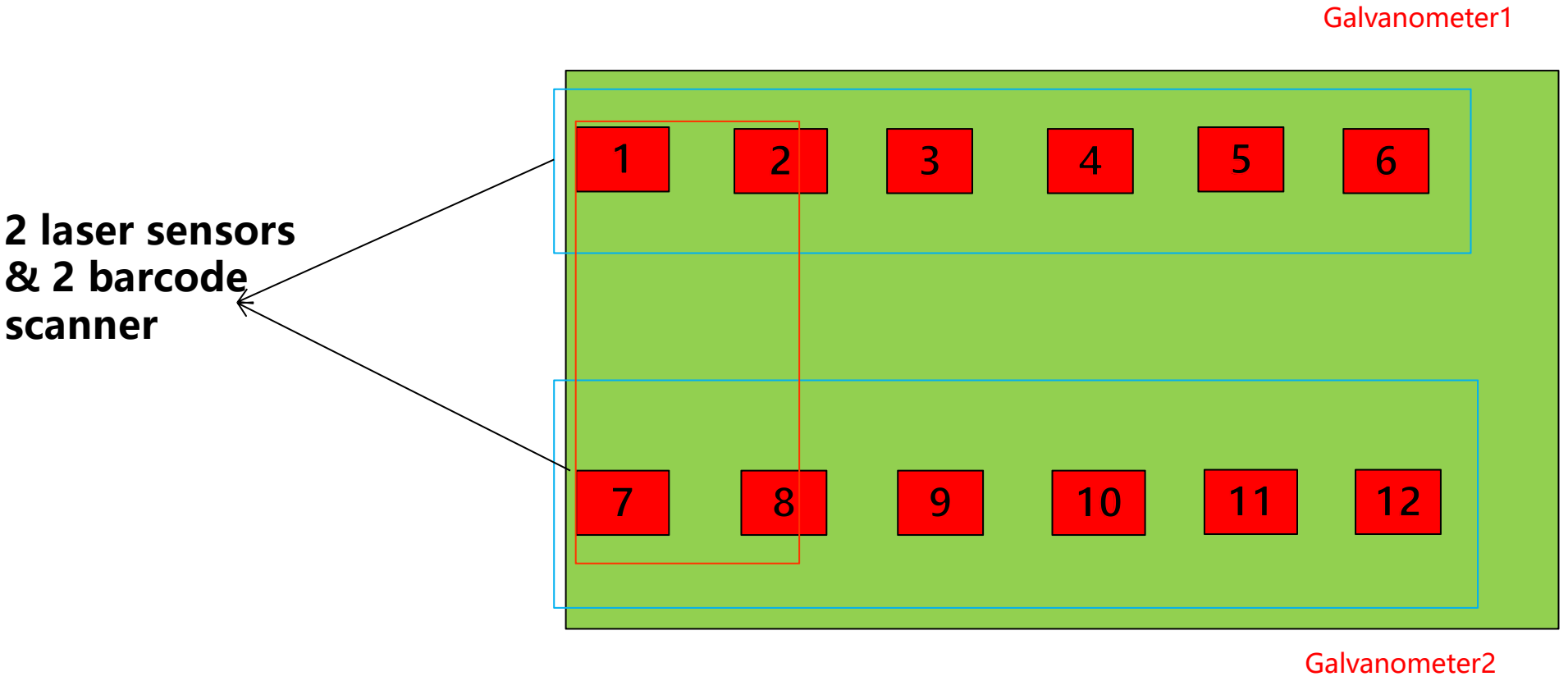
2.4 Barcode scan and height measure station



When Laser sensor measure the height position of product and fixture, QR reading at the same time.

Barcode scan station

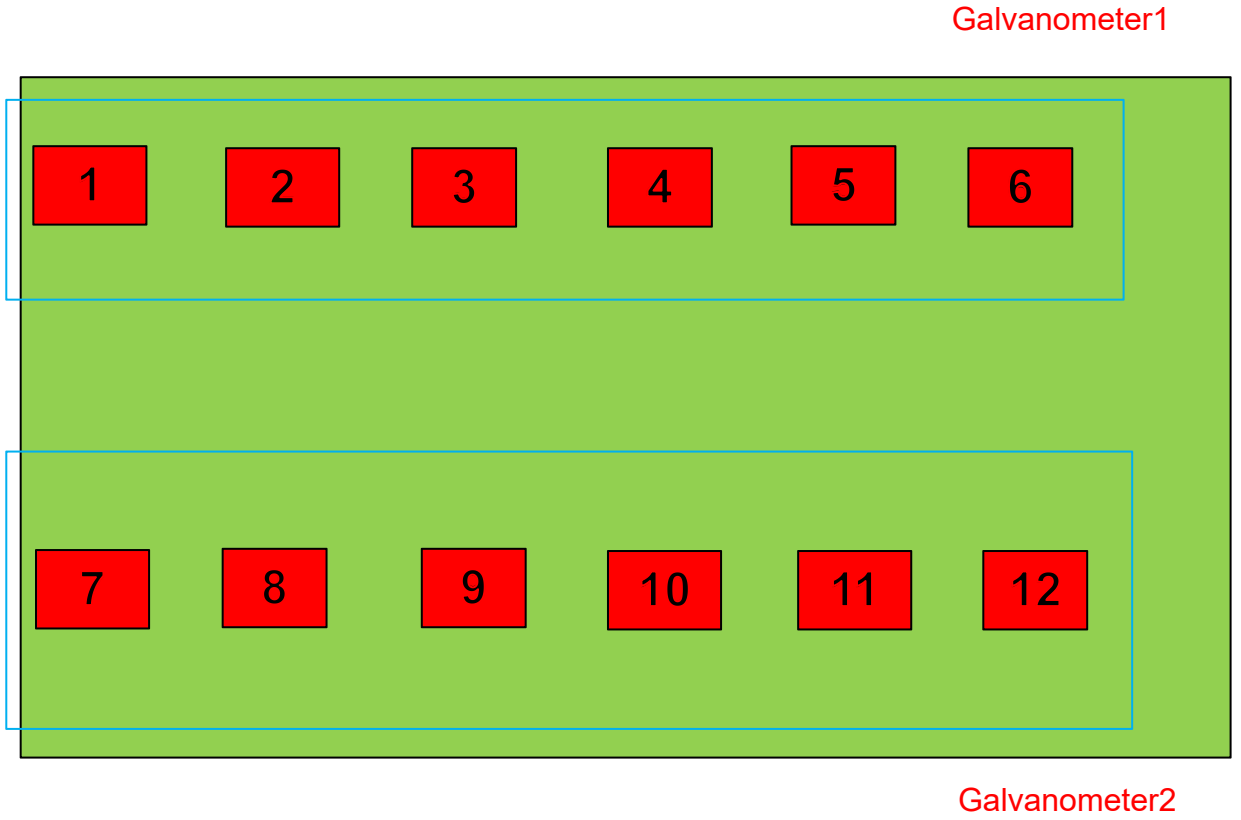
Laser sensors work flow



1&7 scan together，then shift to 2&8, 3&9,4&10, 5&11, 6&12

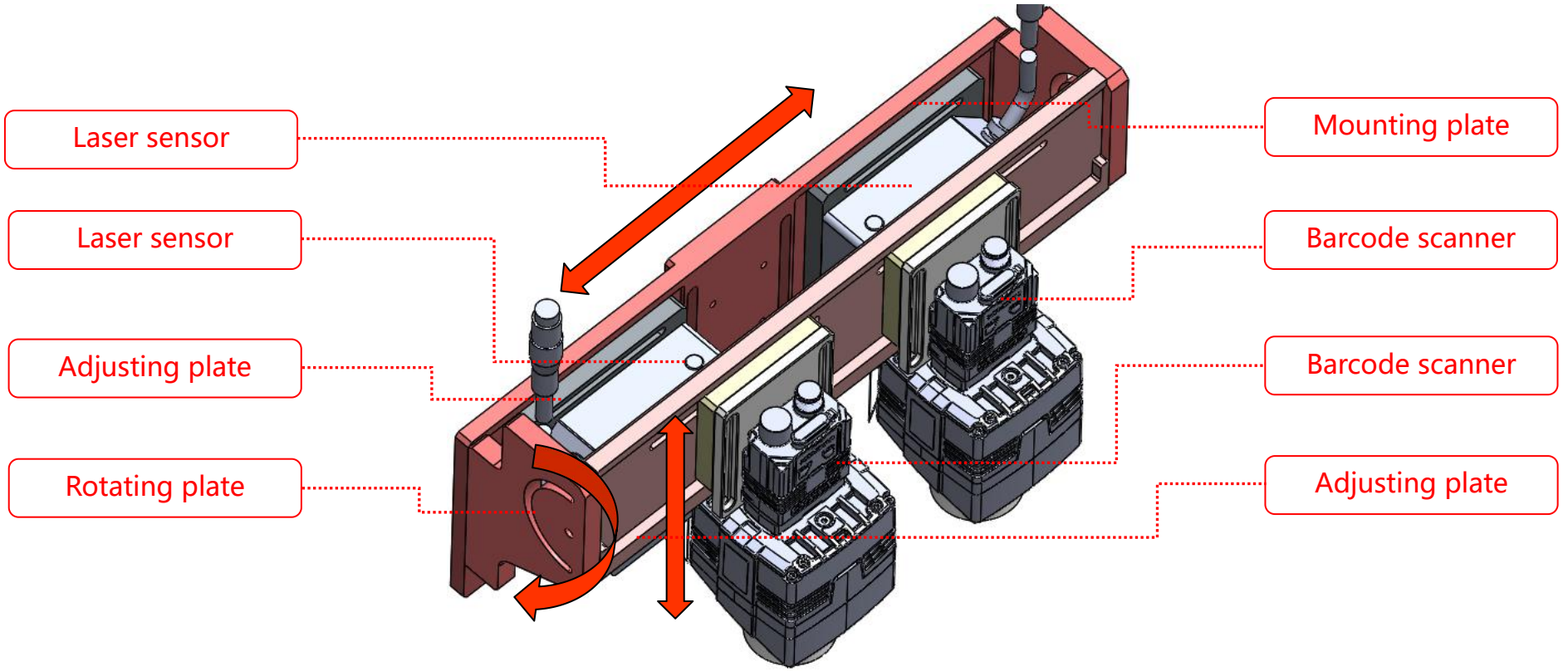
Barcode scan station

Welding work flow



Cavity 1&2 welded together by galva1
Cavity 7&8 welded together by galva2
Then X shift
Cavity 3&4 welded together by galva1
Cavity 9&10 welded together by galva2
Then X shift
Cavity 5&6 welded together by galva1
Cavity 11&12 welded together by galva2

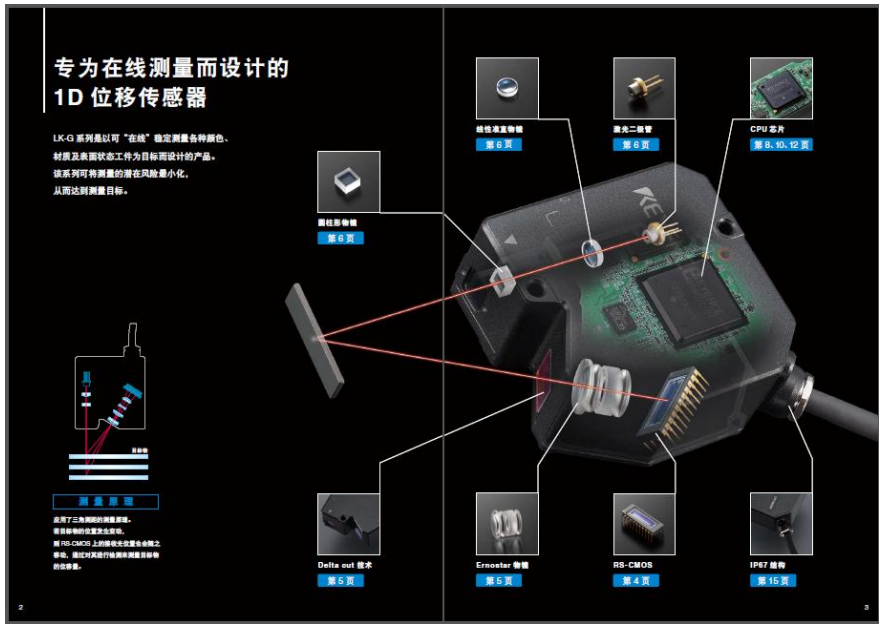
Barcode scanner and height measure



Barcode scanning station

2.4.1 Kernel module

Laser sensor



Type of Laser sensor：
LK-G80

■ LK-G85H/LK-G80H/LK-G87H/LK-G82H

Model		LK-G85H/LK-G80H/LK-G87H/LK-G82H	
Setup mode		Diffuse reflection	Specular reflection
Reference distance		80 mm	75.2 mm
Measurement range *3		±15 mm	±14 mm
Light source		Red semiconductor laser	
	Wavelength	655 nm (visible light) class 3R (IEC60825-1) (G85H/G80H) 655 nm (visible light) class 2 (IEC60825-1) (G87H/G82H)	
	Output	4.8 mW max. (G85H/G80H) 0.95 mW max. (G87H/G82H)	
Spot diameter (at reference distance)		Approx. 70 x 1100 µm (G85/G87) Approx. ϕ70 µm (G80/G82)	
Linearity *1		±0.05% of F.S. (F.S. = ±15 mm)	
Resolution *2		0.3 µm	
Sampling rate		20/50/100/200/500/1000 µs (selectable among 6 steps)	
LED indication		Near the center of measurement range: Lights up green Within the measurement range: Lights up orange Outside the measurement range: Flashes orange	
Temperature characteristics		0.01% of F.S./°C (F.S. = ±15 mm)	
Environ-mental resistance	Enclosure rating	IP67 (IEC60529)	
	Operating ambient light	Incandescent lamp or fluorescent lamp: 10000 lux max.	
	Operating ambient temperature	0 to +50 °C	
	Operating ambient humidity	35 to 85% RH (No condensation)	
	Vibration resistance	10 to 55 Hz, double amplitude 1.5 mm 2 hours in each direction X, Y, Z	
Material		Aluminum die-cast	
Weight (including cable)		Approx. 380g	

*1 The value when measuring the KEYENCE standard target (ceramic) in standard mode
*2 The averaged value when measuring the KEYENCE standard target (SUS) 4096 times at the reference distance.
*3 The value when measuring the KEYENCE standard target (ceramic). When the sampling rate is 20 µs, the measurement range is between –9 mm (FAR side) and –15 mm (NEAR side) for diffuse reflection and between –8.7 mm (FAR side) and –14 mm (NEAR side) for specular reflection.

2.4.2 Kernel module

Barcode scanner



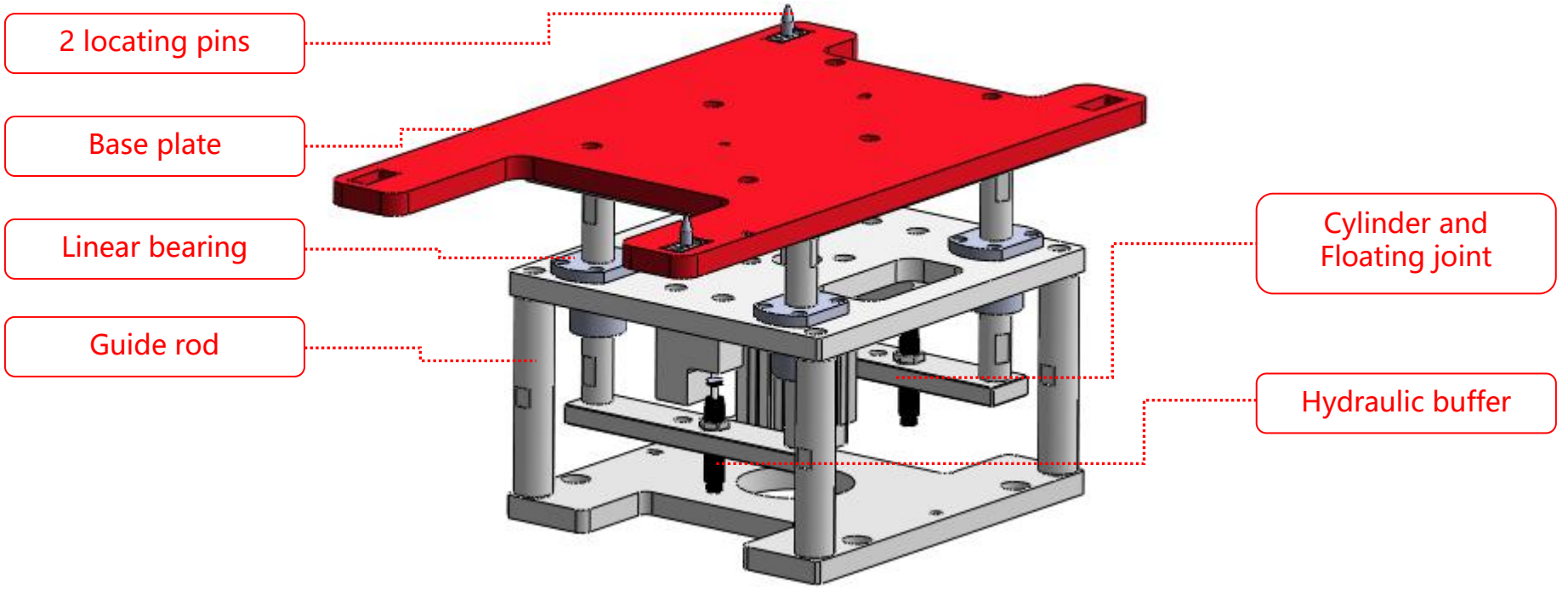
Type of Bar code scanner：
DataMan 282

DATAMAN 280 SERIES SPECIFICATIONS

	DataMan 280	DataMan 282
Algorithms and Technologies	1DMax, 2DMax, Hotbars, PowerGrid	
Image Sensor	1/3" CMOS	1/2.8" CMOS
Image Sensor Properties	Diagonal 6.21 mm, 3.45 µm square pixels	Diagonal 6.17 mm, 2.8 µm square pixels
Image Resolution	1440 x 1080 pixels	1920 x 1080 pixels
Electronic Shutter Speed	Min. exposure: 29 µs Max. exposure: Up to 10 ms (internal illumination) / Up to 200 ms (external illumination)	
Acquisition	Up to 45 Hz	
Lens Options	6.2 mm or 16 mm High Speed Liquid Lens	8 mm, 12 mm, or 16 mm High Speed Liquid Lens
Trigger and Tune Buttons	Yes, with Quick Setup Intelligent Tuning	
Aimer	2 LEDs	Laser cross pattern (class 2)
Discrete Inputs	2 opto-isolated, 2 configurable	
Discrete Outputs	2 opto-isolated, 2 configurable	
Other I/O Points	2 user configurable as inputs or outputs	
Status Outputs	5 status LEDs, audible beeper, 2 visual indicator lights	5 status LEDs, audible beeper, 2 visual indicator lights, 4 HPIT light color options
Lighting	Standard: 4 LED lights for 6.2mm lens (red, blue, white, IR, and UV options) HPIL: 4 LED high-powered lights for 16 mm lens (red, white options) Optional bandpass filters. Front cover options: polarized/half-polarized and clear.	HPIT: 4 multi-color LED lights (red, green, blue, and white) Optional bandpass filters. Front cover options: clear, diffused, polarized and dome.
Communications	Serial and 1 GB/second Ethernet interface, USB-C, and keyboard capability	
Protocols	RS-232, TCP/IP, PROFINET (class B), EtherNet/IP™, SLMP, CC-Link, Modbus TCP, NTP, SFTP, FTP, MRS, Java Scripting enabled for custom protocols	
Power	24V +/- 10% and Power over Ethernet (PoE)	24V +/- 10%
Power Consumption	≤7.5W	
Material	Zinc alloy/aluminum	
Weight	6.2 mm: 141 g; 16 mm: 169 g Right angle configuration adds 50 g	290 g Right angle configuration adds 50 g
Operating Temperature	0–40° C (32–104° F)	
Storage Temperature	-10–60° C (14–140° F)	
Operating and Storage Humidity	<95% non-condensing	
Protection	IP67	
RoHS Certified	Yes	
Approvals	EU [CE], US [FCC], TUV, CB, NRTL, IEC 61010, Korea [KCC], India [BIS]	

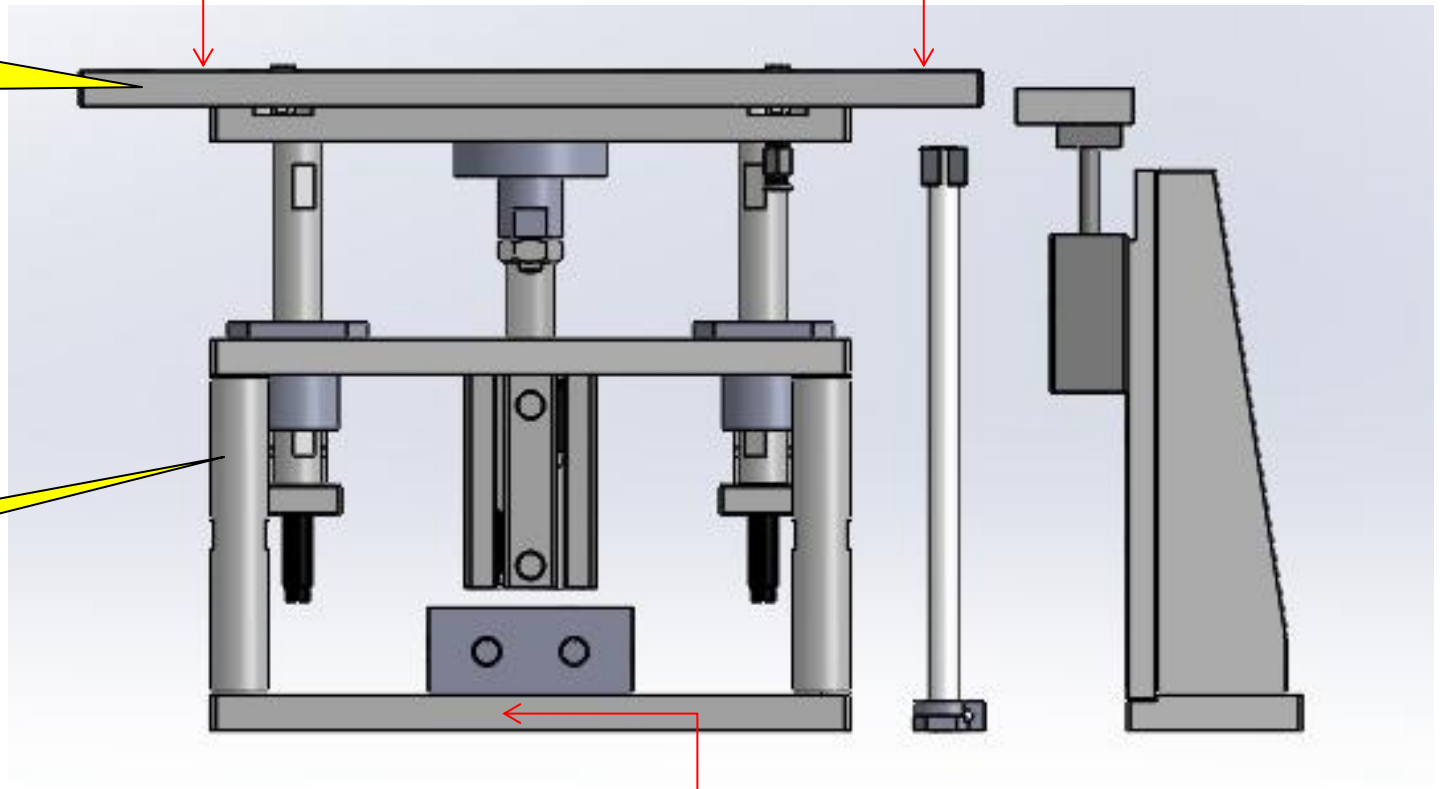
Lift and Locate Pallet

Main supplier
Cylinders and accessories:Airtac
Guide rod and Linear bearing:Yiheda
Proximity sensor:Banner



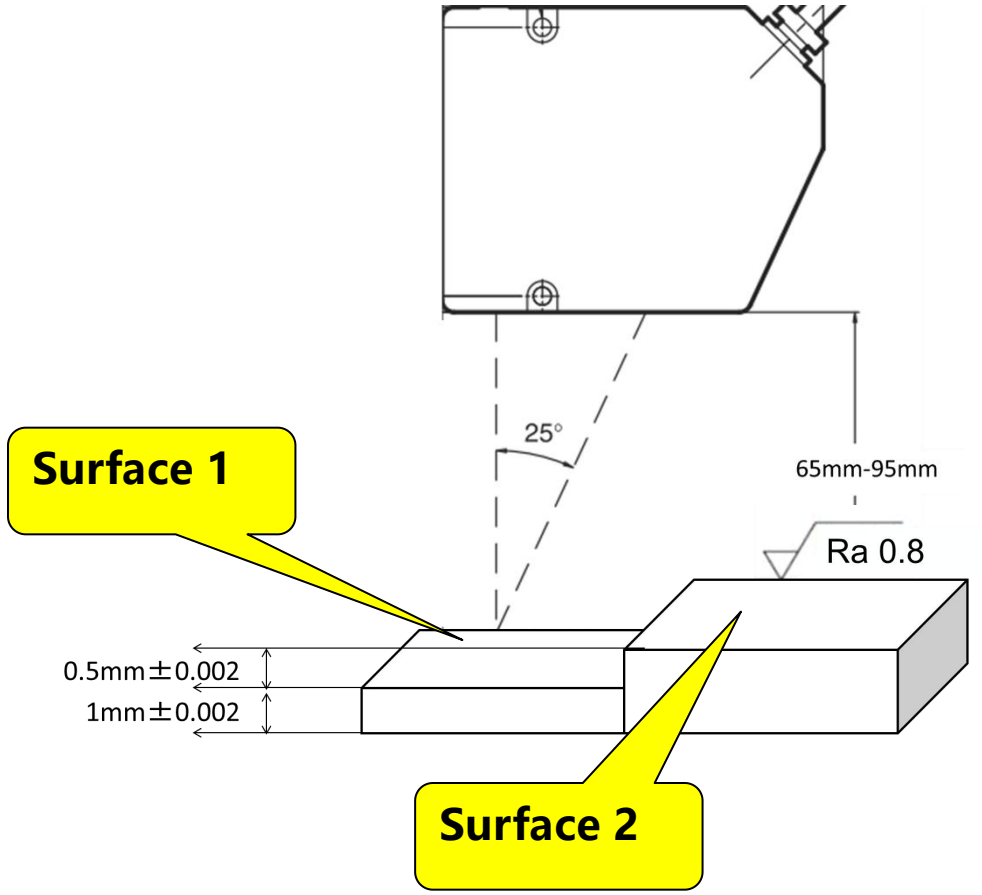
2.4.4 Barcode scan and height measure station

Lift and Locate Pallet



2.5.1 Calibration the Laser sensor by height block

1.Set the calibration block on baseplate
2.move to measure the surface1
3.move to measure the surface2.
4.Compare the distance of surface1 and surface 2，ΔH=0.5±0.002



Calibration Block

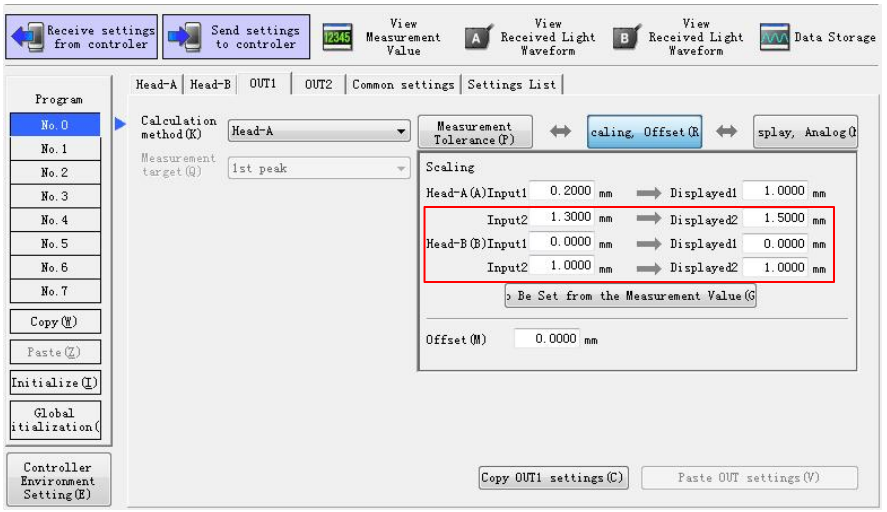
2.5.2 Calibration of the Laser sensor by height block

Laser Setup Procedure

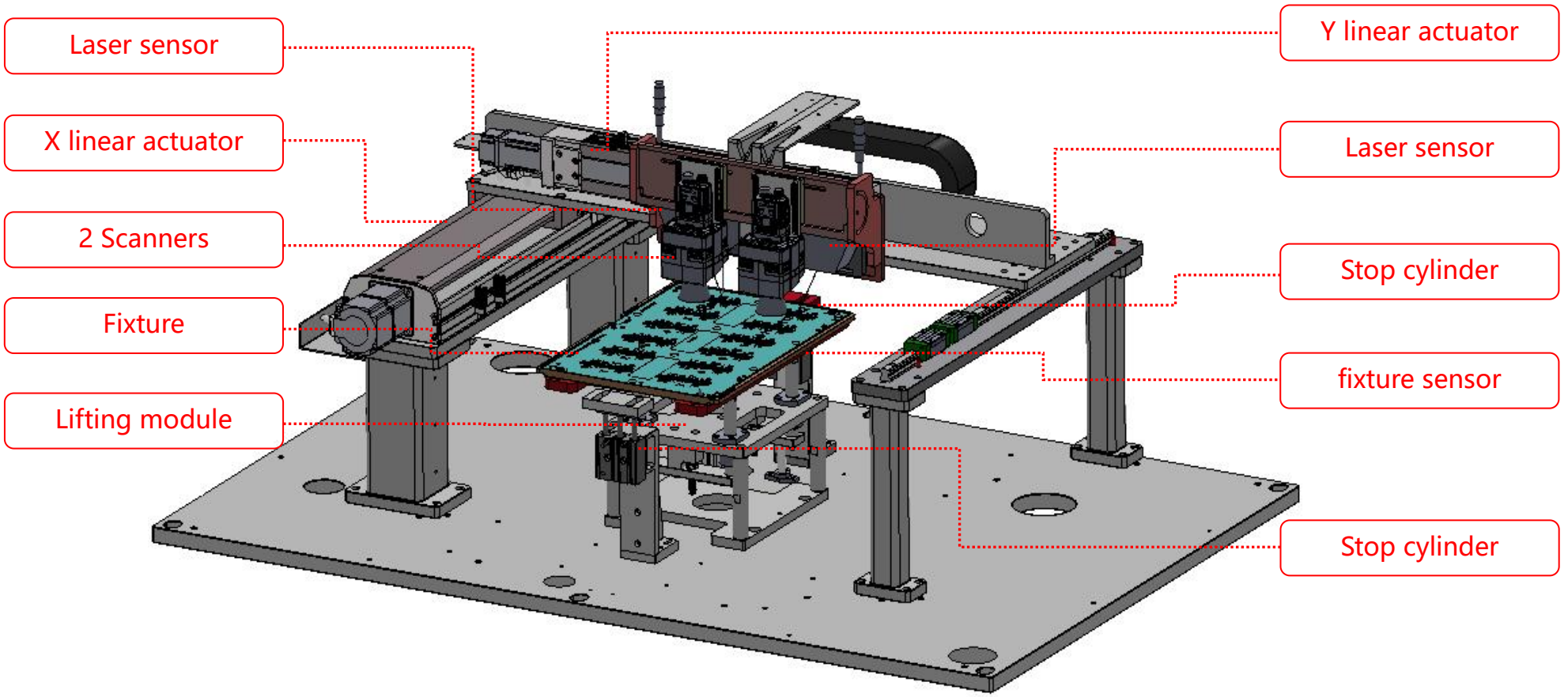
- **Calibration**

Confirm the laser measurement value with gauge block. If the measurement value have error, correct the value with gauge bock.

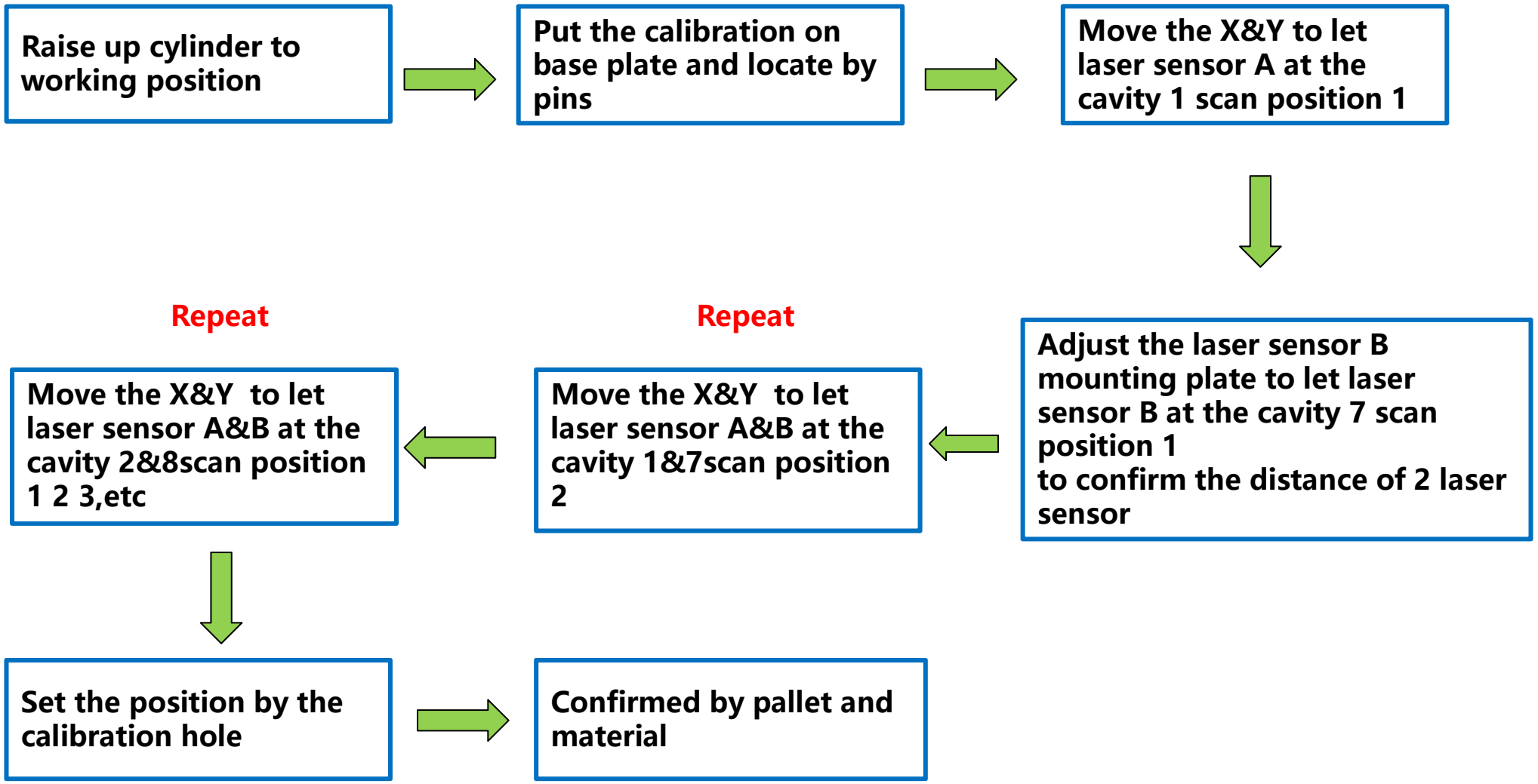
Ex.) Setting scaling for OUT1 with 1 mm and 1.5 mm gauge blocks
(1) Click "Set from measurement data" in the scaling/offset settings for OUT1.
(2) Insert the 1 mm gauge, click "Set" for input value 1 and then enter 1 mm for display value 1.
(3) Insert the 1.5 mm gauge, click "Set" for input value 2 and then enter 1.5 mm for display value 2.
(4) Click "Send settings to controller".



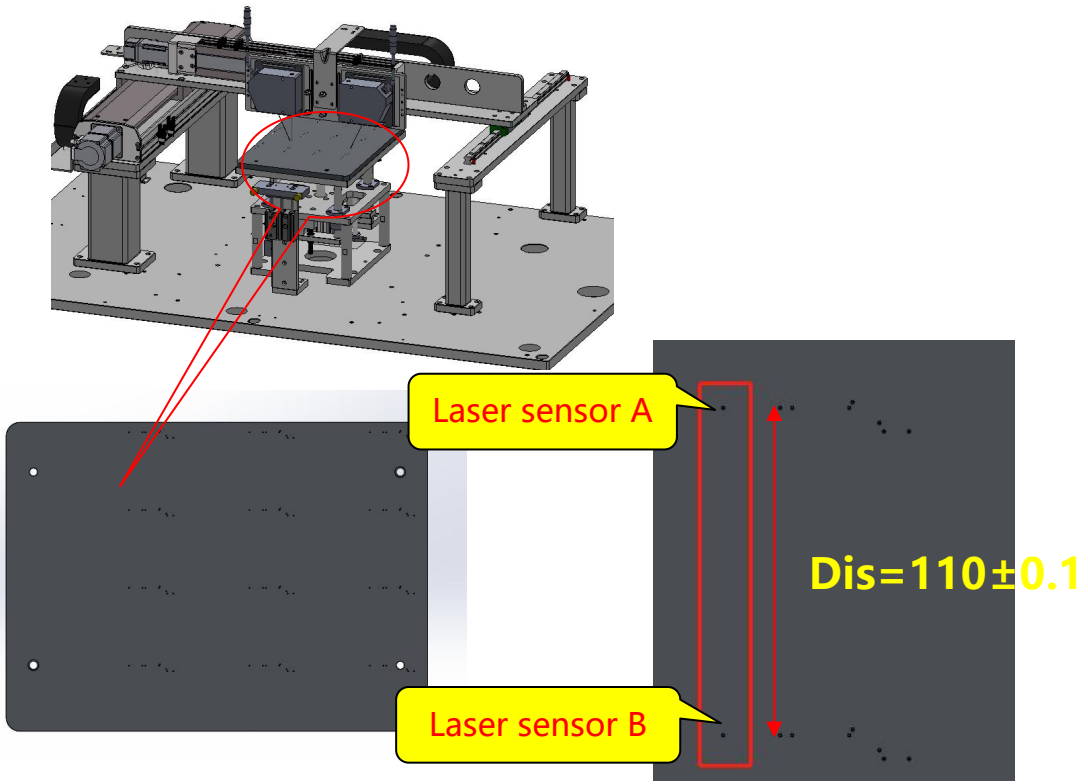
2.5.2 Measuring station set up and calibration

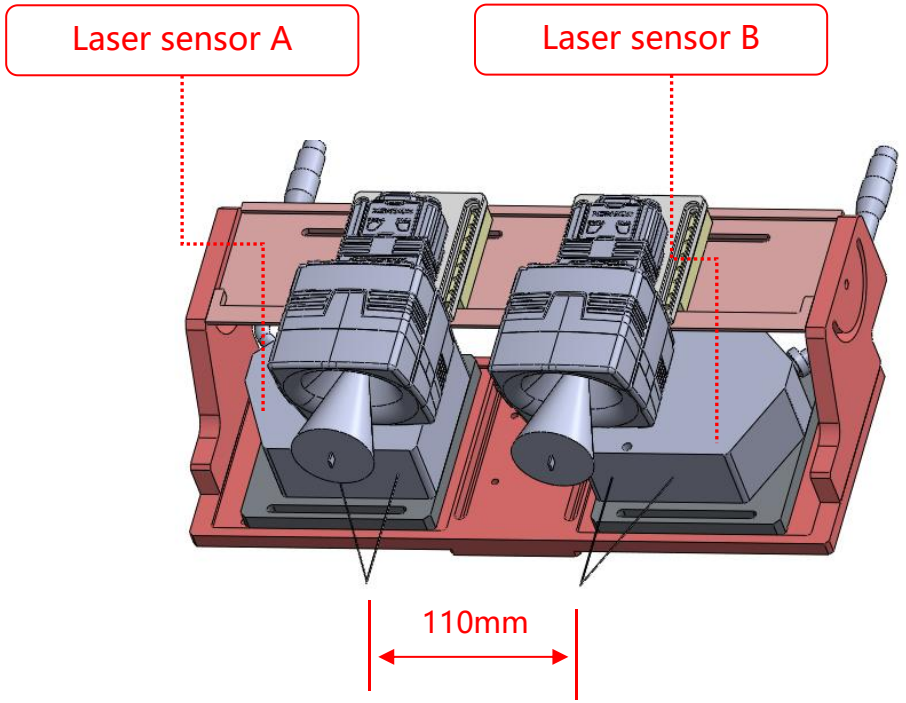


2.5.3 Scanning flow



2.5.4 Calibration the distance of Laser sensor and Angel

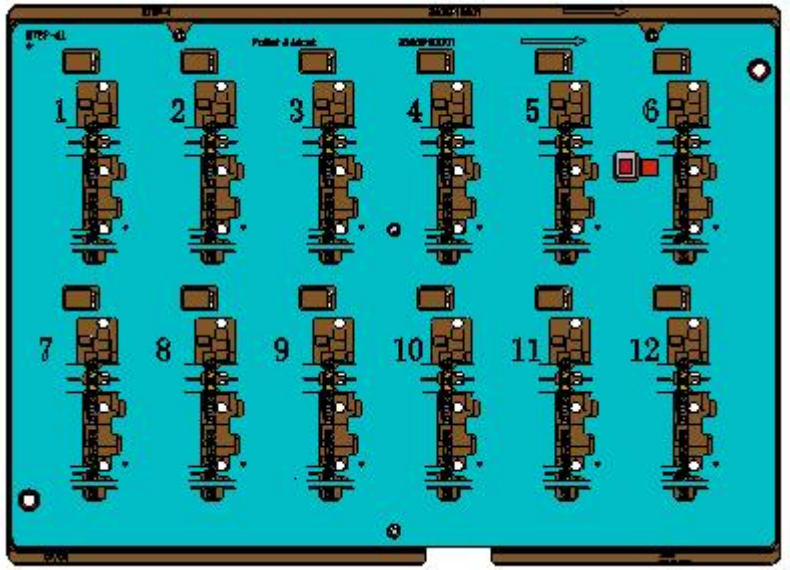


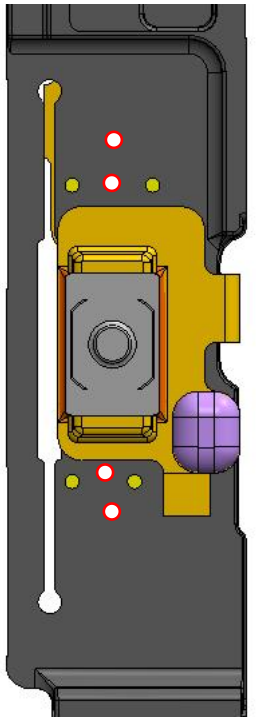


1.Locate the test block on base plate and cylinder raise to working position.
2.Move the X Y to confirm the light of laser sensor A is in hole.
3.Adjust the laser sensor B mounting plate until the light in the hole too.
4.Fasten the screws and mark line.

2.5.5 FBW Measure points

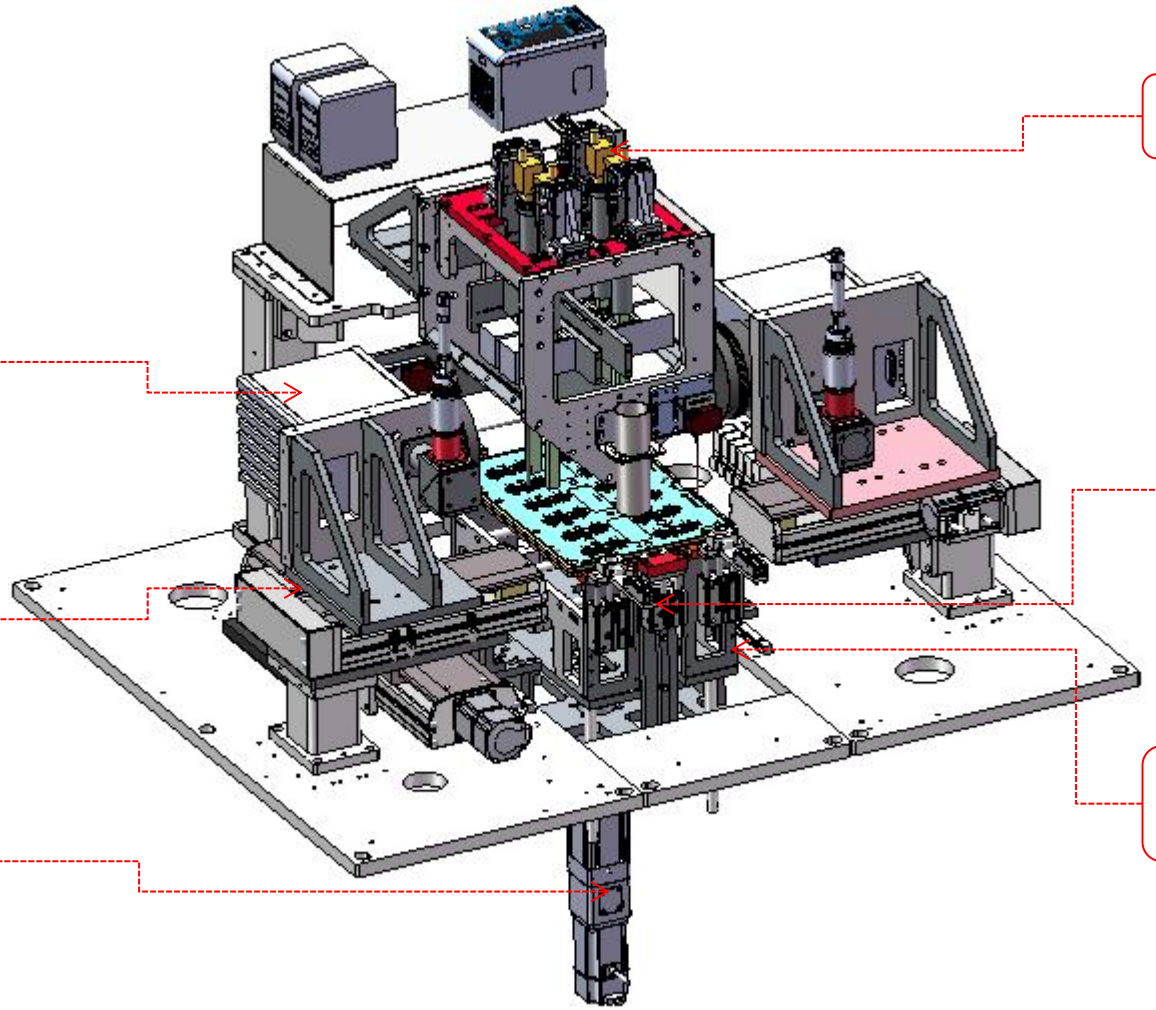
2 laser sensors





height points:
2 points of Volume bracket
2 points of welding area
ΔH spec 0-0.17mm

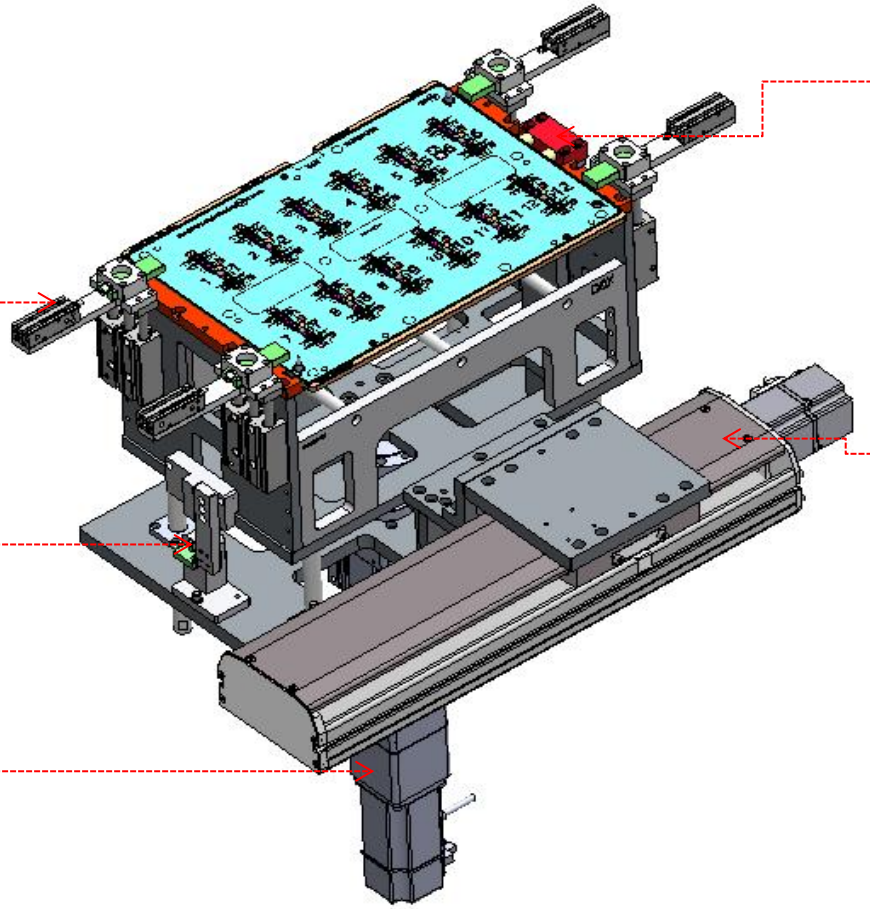
2.6 Auto welding station



Auto welding station

2.6.1 Auto welding station

Pallet Lifting and Locating mechanism



Auto welding station

2.6.2 Kernel module

Lifting and locating mechanism electric actuator

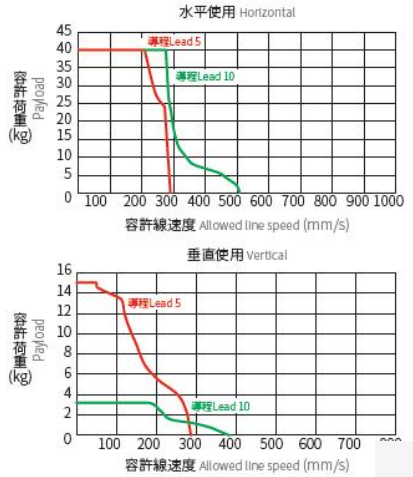
基本仕様 Specification

規格 Spec	位置重複精度 Repeatability (mm)		±0.01	
	螺桿導程 Ball Screw Lead (mm)		5	10
	最高速度 Maximum Speed (mm/s)※2	水平使用 Horizontal	≤288	≤508
		垂直使用 Vertical	≤250	≤418
	最大可搬重量 Maximum Payload	水平使用 Horizontal (kg)	≤40	≤40
		垂直使用 Vertical (kg)	≤14.7	≤2.9
	定格推力 Rated Thrust (N)		565	283
	標準行程 Stroke / Pitch (mm)※1		50-300 / 50	
部品 Parts	馬達尺寸 Motor Dimension (mm)		42□	
	滾珠螺桿規格 Ball Screw Spec		C7 φ12mm	

※1 行程超過200時，會產生螺桿偏擺震動，此時請將速度調降。詳細請參考P45頁說明。
When the stroke is over 200mm, the run-out of the ballscrew will occur.
We recommend to low down the working speed under this circumstances.Please refer to P45 for details.

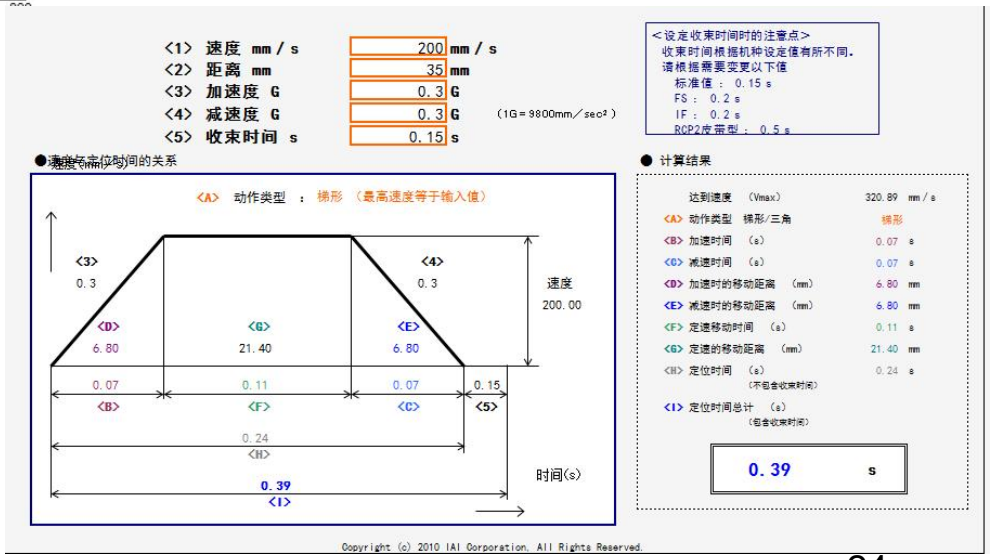
※2 此處最高速度為軟體移動速度設定為100%時。
The maximum speed shown here is when software speed setting is 100%.

速度-荷重曲線圖 Speed-Payload Relationship

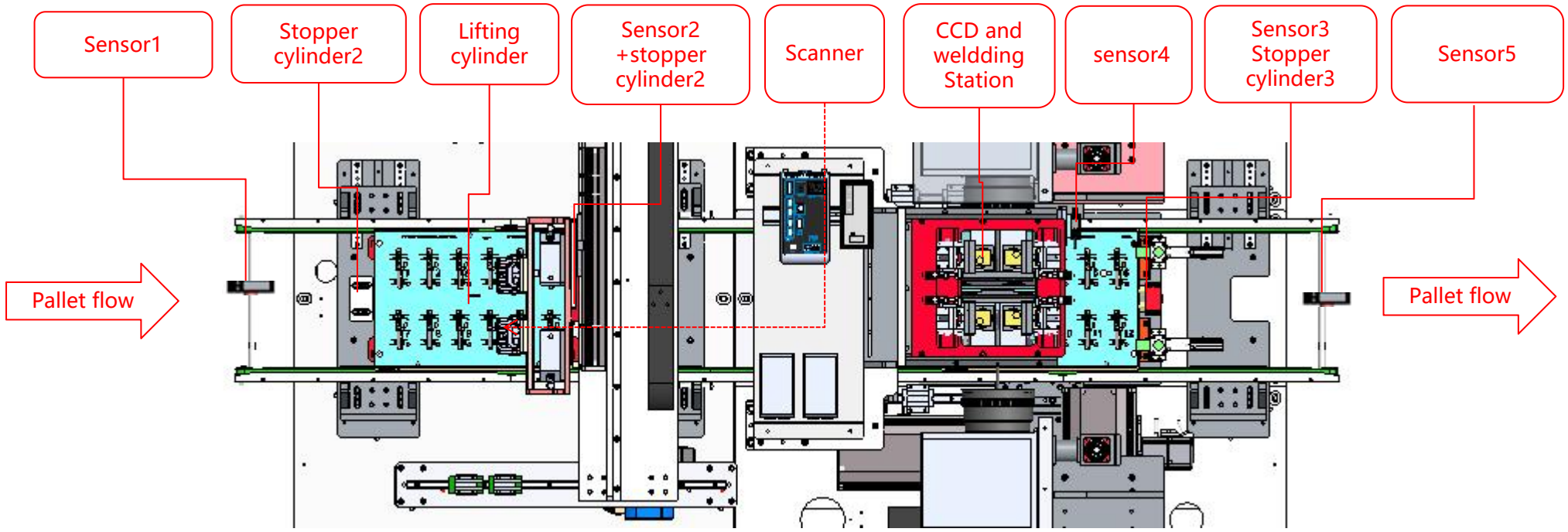




S=35mm,Vmax=200mm/s, t=0.39s

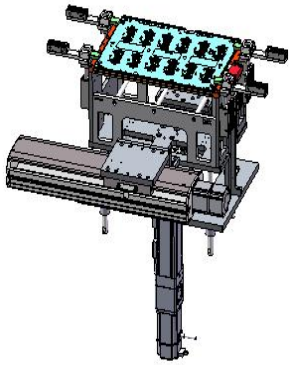


2 .7 Pallet Flow

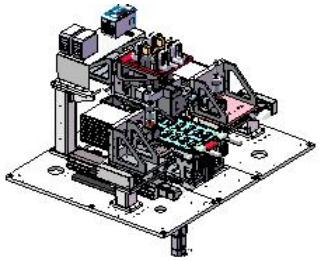


Sensor1 off machine will ask for pallet from previous machine → Sensor2 detect pallet and lifting cylinder raise → X1 Y1 move to read barcode and measure heightl → After finish measuring and scanning, lifting cylinder drop → Pallet flow to CCD and welding station → Sensor3 detected pallet, Stopper cylinder3 raised → Z actuator raise and clamp carrier → sensor 4 detected pallet is Level, CCD will take picture and welding → Sensor 5 off and posterior machine ask for pallet, stopper cylinder3 drop

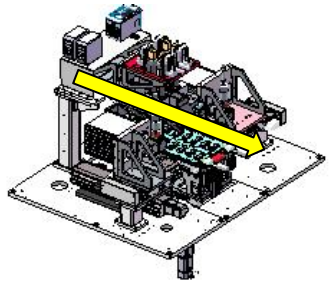
2.8 Welding working Flow



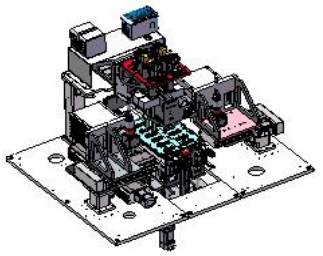
Sensor get single when pallet stopped by cylinder, electric actuator rise up and locate pallet, then clamping cylinders clamp pallet .



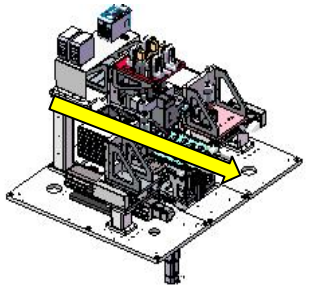
Cavity #1,#2,#7&#8, 4 CCDs take picture for locating,then laser welding,CCD inspection



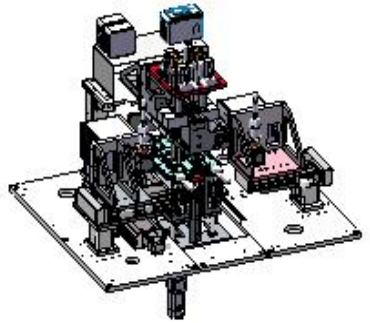
X-actuator move to Cavity #3,#4,#9&#10



Cavity #3,#4,#9&#10, 4 CCDs take picture for locating,then laser welding,CCD inspection



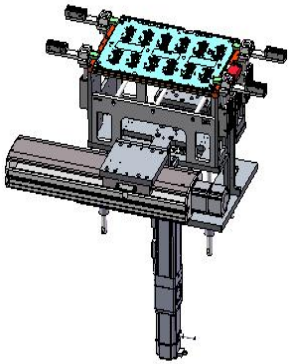
X-actuator move to Cavity #5,#6,#11&#12



Cavity #5,#6,#11&#12 4 CCDs take picture for locating,then laser welding,CCD inspection

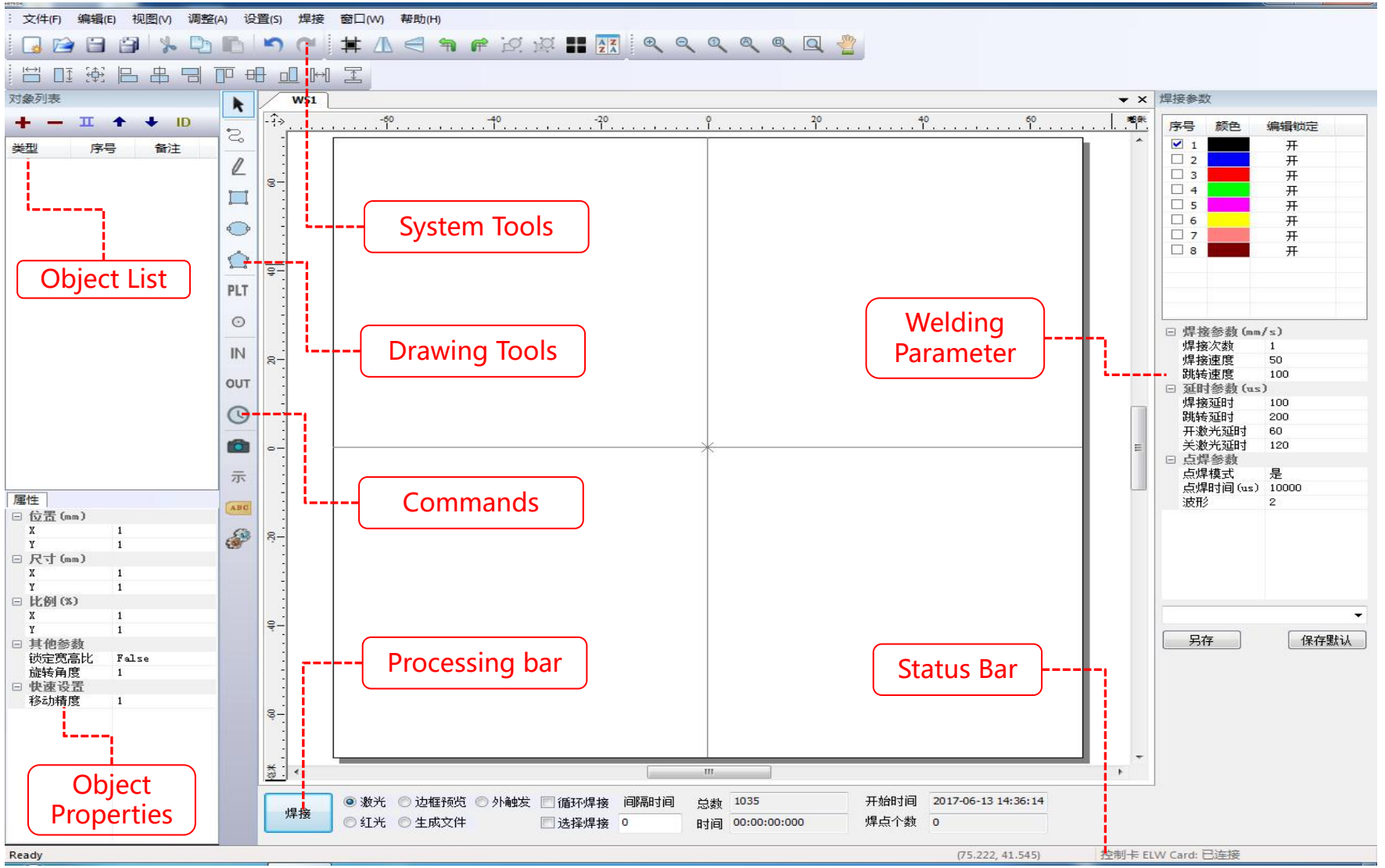


Upload SFC&PDCA



electric actuator drop to zero position and clamping cylinder unclinch pallet, then stopper cylinder let it go.

Software control

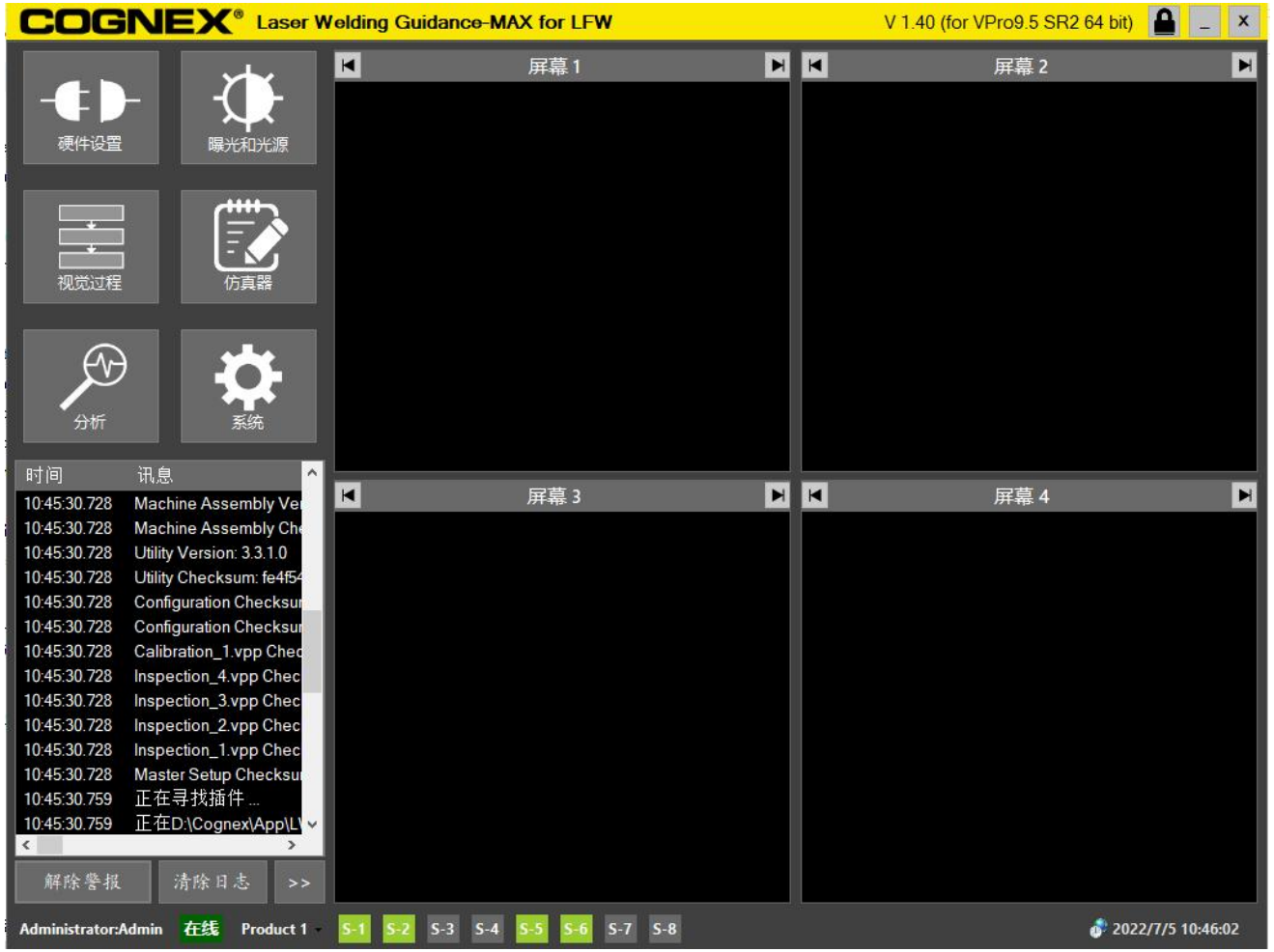


2.9 Visual control system

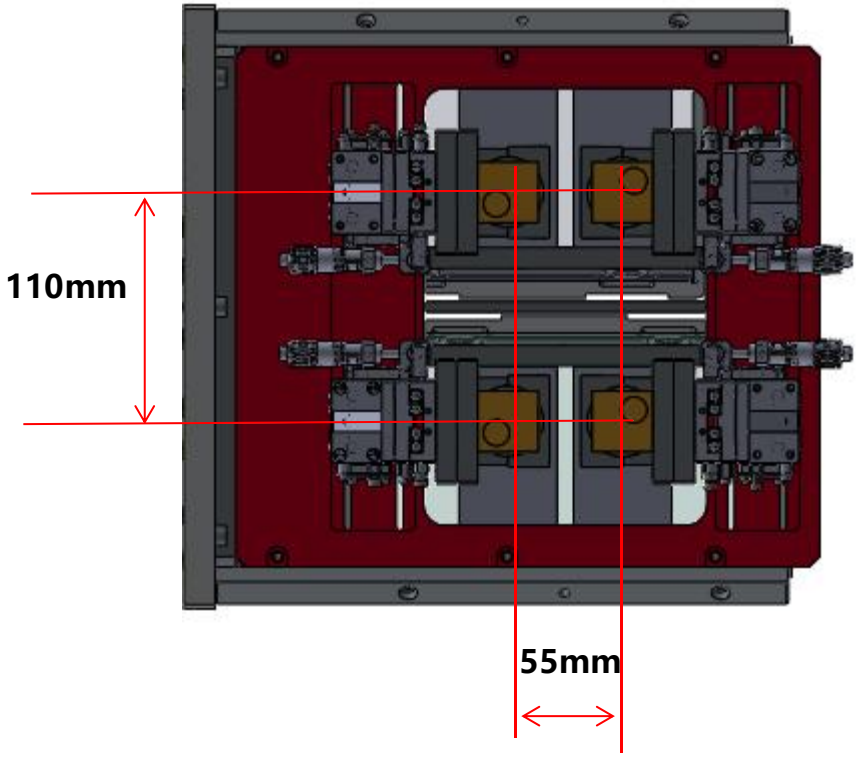
CCD system work flow

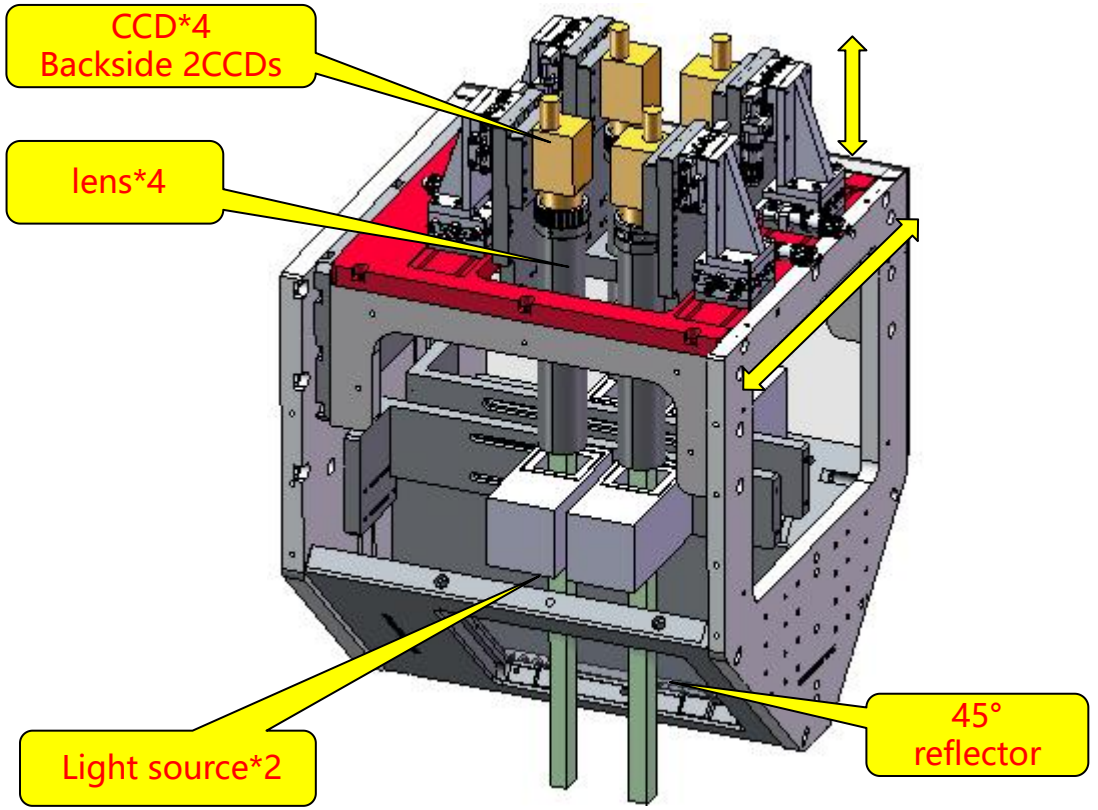
See vision flow from Cognex

LWG APP



External Coaxial Optical System





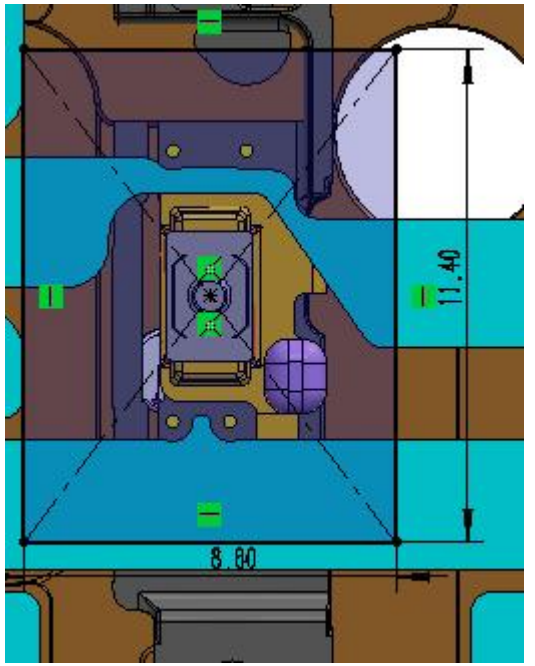
3.0 Kernel module

Vision accuracy

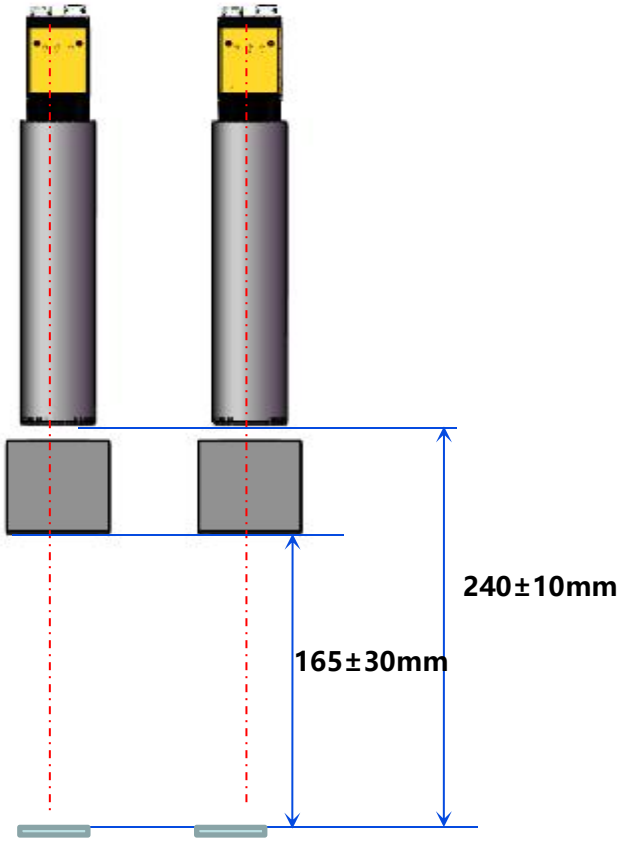
The CCD hardware will be using Cognex USA.

Station	CCD	Resolution	FOV	Accuracy
RFW	Cognex: CAM-CIC-5000-14-G	2456*2058	11.4mm*8.6mm	0.0044mm/pixel

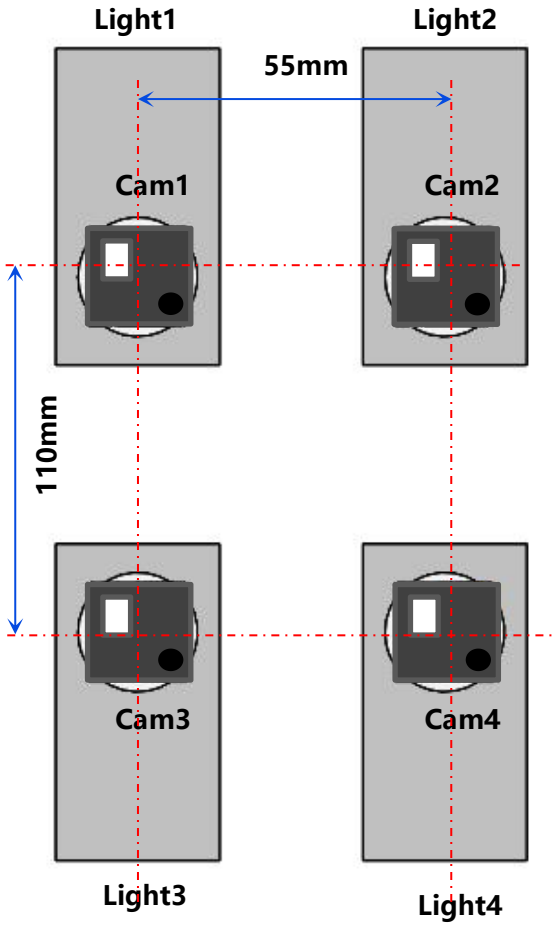
FBW		数量
CAM1,2,3,4	Cognex CAM-CIC-5000R-14-G	4
Lens1,2,3,4	Telecentric Lens	
Lens Model	OPT-AHL05-240-01	4
Working Distance	240±10mm	
FOV	11.4mm*8.6mm	
Spatial Resolution	0.0044mm/pixel	
Aperture	/	
Extension Tube	/	
Light Source	Red Coaxial Light	
Light Model	OPT-CTD50-R	4



External Coaxial Optical System

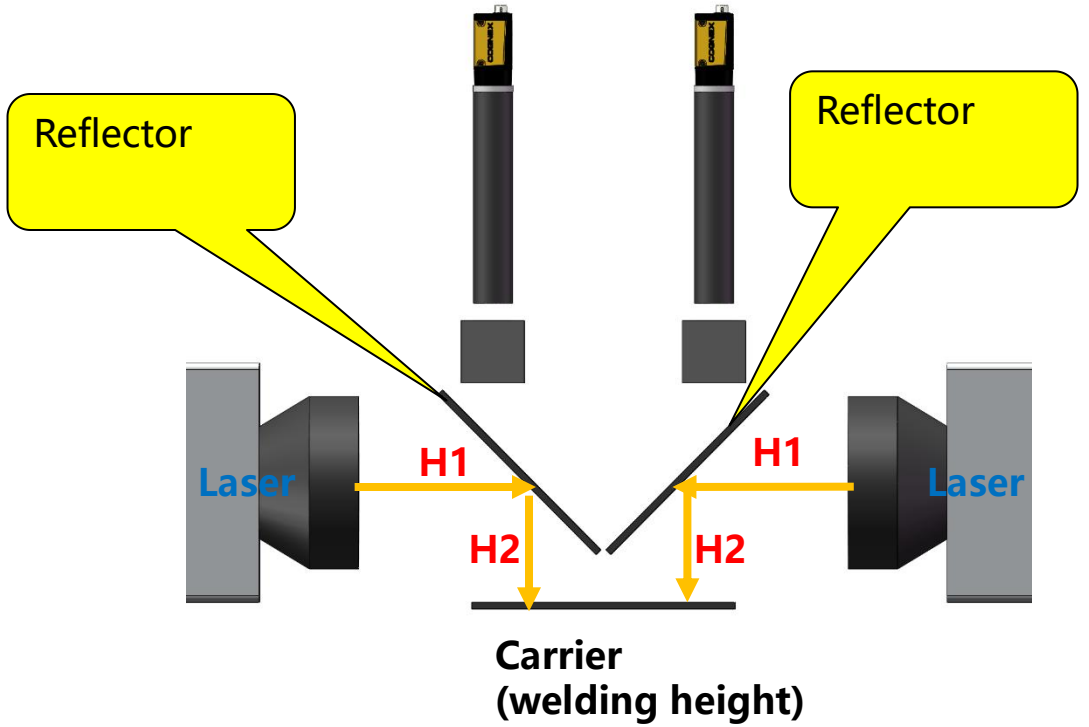


光源示意图正视图
Schematic diagram of setup (Front View)



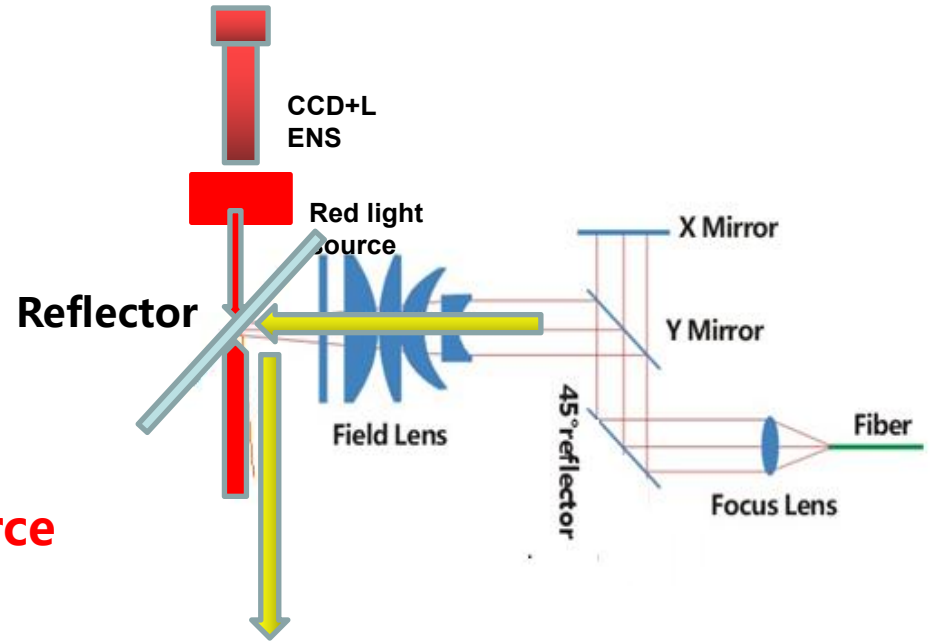
光源示意图俯视图
Schematic diagram of setup (Top View)

Laser work distance



Laser work distance=H1+H2
H1 control by Laser head actuator
H2 control by Z actuator and pallet level

Optical System



Light source

Optical Layout

Components List of cylinder and actuator

NO	Station	Components	Description	Qty	Supplier
1	Barcode reading station	X1 linear actuator	Travel Distance: 350mm, 400W servo motor	1	HG
2		Y1 linear actuator	Travel Distance: 250mm 100W servo motor	1	HG
3		Lifting cylinder	ACQ32X40SB	1	Airtac
4		Stopper cylinder	TCM12x25S	2	Airtac
5	Welding station	X2 linear actuator	Travel Distance: 300mm, 400W servo motor	1	HG
6		Z linear actuator	Travel Distance: 50mm 400W servo motor with brake	1	HG
7		Y2 linear actuator	Travel Distance: 100mm, 200W servo motor	2	HG
8		Clamping cylinder	ACQ12X35S	4	Airtac
9		Clamping cylinder	MD6x25S	4	Airtac
10		Stopper cylinder	TCM12x20S	1	Airtac

Components List of Optical System

NO	Components	Description
1	Laser Source	QCW150（IPG）
2	CCD	5 mega-pixel
3	Galvanometer Head	Φ30
4	Focus Lens	F75
5	Lens	F210

Laser Source



Type of Laser Source： QCW150

Parameter	Parameter value
Specification	IPG,QCW150
Center wavelength	1064nm
Peak power	1500w
Power unstability	≤3%

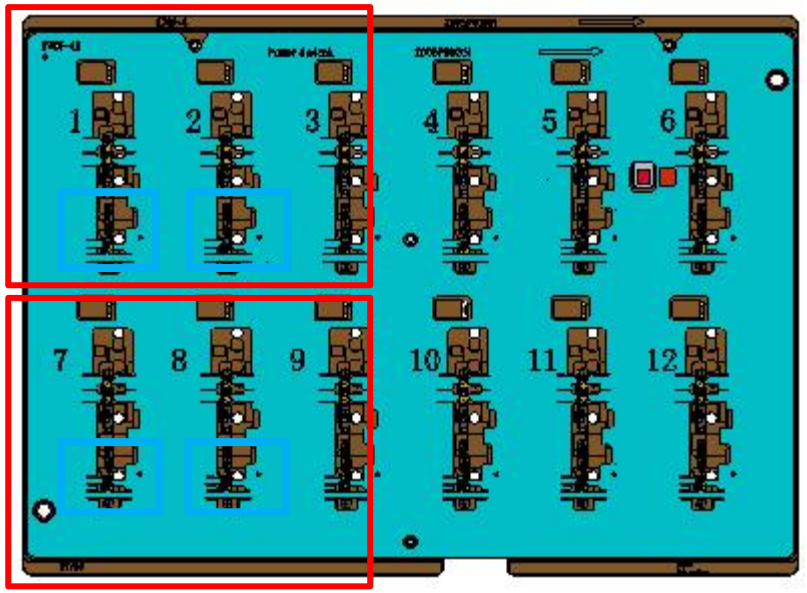
Galvanometer Head



Parameter	Parameter value
Aperture size	Φ30
Repeatability	< 8μRad
Marking speed	3m/s
Positioning speed	8m/s

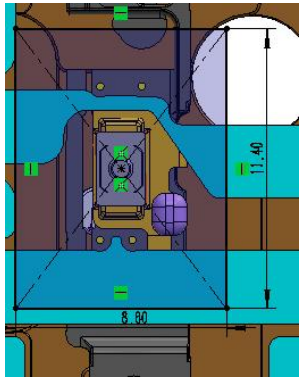
4 Tractor parameter

NO	Components	Description	Remarks
1	Energy stability of the whole machine	≤3%	
2	Effective welding width	110*110mm	
3	The effective width error precision	±0.5mm	
4	Z-axis Automation range	50mm	
5	Monitor	17 inch	
6	Cool system	Air cooled	
7	Working temperature /humidity	10~35℃，≤70%	
8	Power	200-240 VAC, 50/60 Hz	



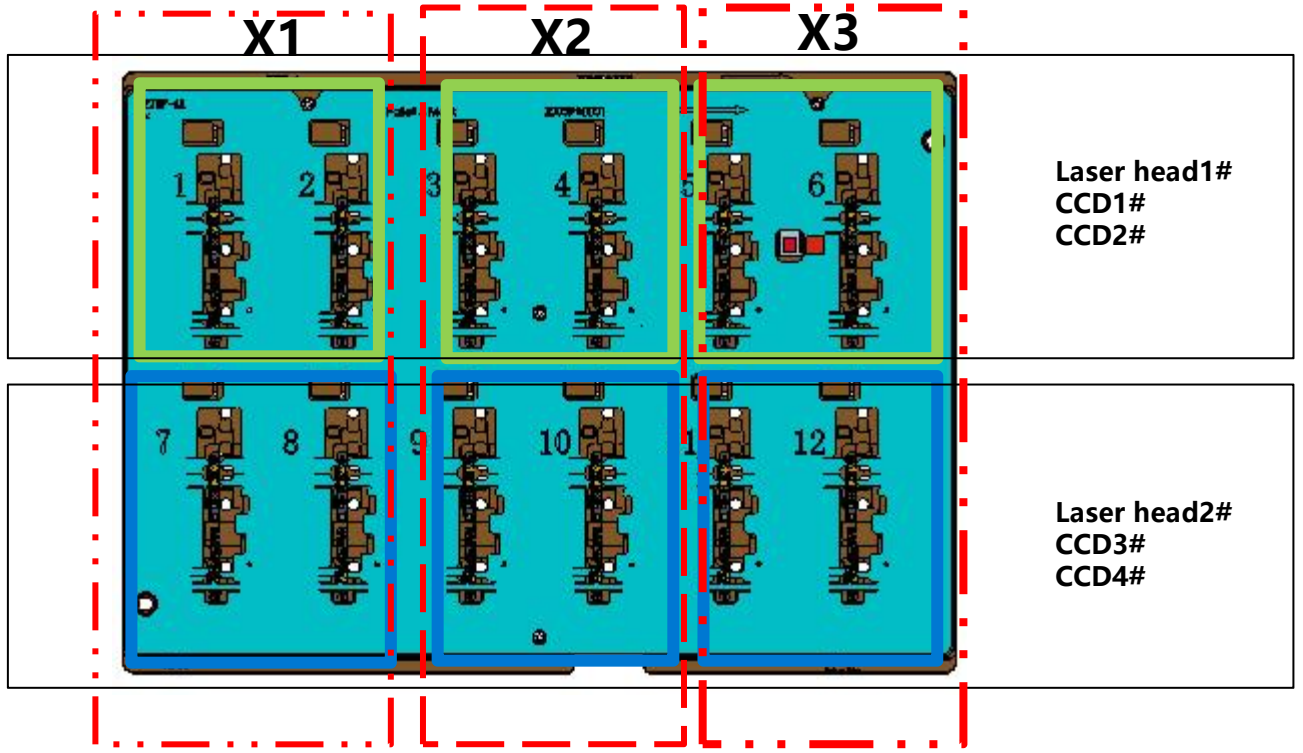
Red block:
Laser1# FOV=110*110mm
Blue block:
CCD1# FOV=11.4*8.6mm
CCD2# FOV=11.4*8.6mm

Red block:
Laser2# FOV=110*110mm
Blue block:
CCD3# FOV=11.4*8.6mm
CCD4# FOV=11.4*8.6mm



CCD FOV image

5 Laser FOV&CCD FOV



Pallet lifted and located

CCD1 locating Cavity1#
CCD2 locating Cavity2#
CCD3 locating Cavity7#
CCD4 locating Cavity8#

Laser head1# welding cavity1#&2#
Laser head2# welding cavity7#&8#

CCD1 recheck Cavity1#
CCD2 recheck Cavity2#
CCD3 recheck Cavity7#
CCD4 recheck Cavity8#

X-axis move to X2 position

CCD1 locating Cavity3#
CCD2 locating Cavity4#
CCD3 locating Cavity9#
CCD4 locating Cavity10#

Laser head1# welding cavity3#&4#
Laser head2# welding cavity9#&10#

CCD1 recheck Cavity3#
CCD2 recheck Cavity4#
CCD3 recheck Cavity9#
CCD4 recheck Cavity10#

X-axis move to X3 position

CCD1 locating Cavity5#
CCD2 locating Cavity6#
CCD3 locating Cavity11#
CCD4 locating Cavity12#

Laser head1# welding cavity5#&6#
Laser head2# welding cavity11#&12#

CCD1 recheck Cavity5#
CCD2 recheck Cavity6#
CCD3 recheck Cavity11#
CCD4 recheck Cavity12#

6 CT chart

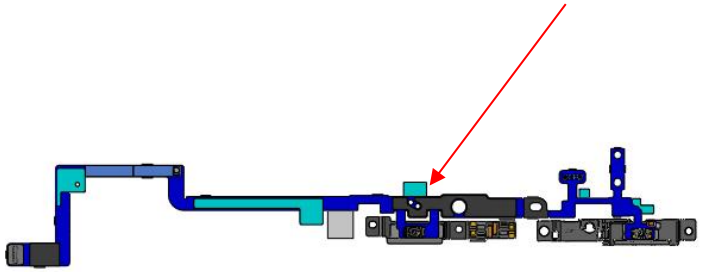
Measuring CT(s)	27.5
Conveyor(s)	4
welding CT(s)	20.9
total CT(s)	27.5+4=31.5
UPH	1304



HG-BFS CT breakdown.xls

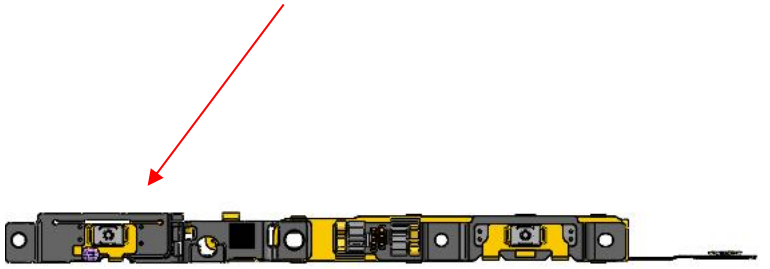
7 D47 EVT Product To V57 P1 Product Comparison

D47-EVT



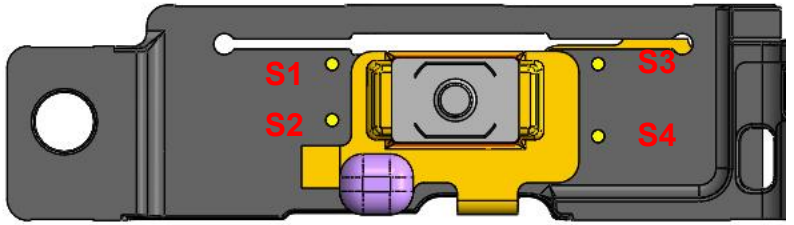
Spot 1&2 SUS(0.1) + SUS(0.28)

V57-P1



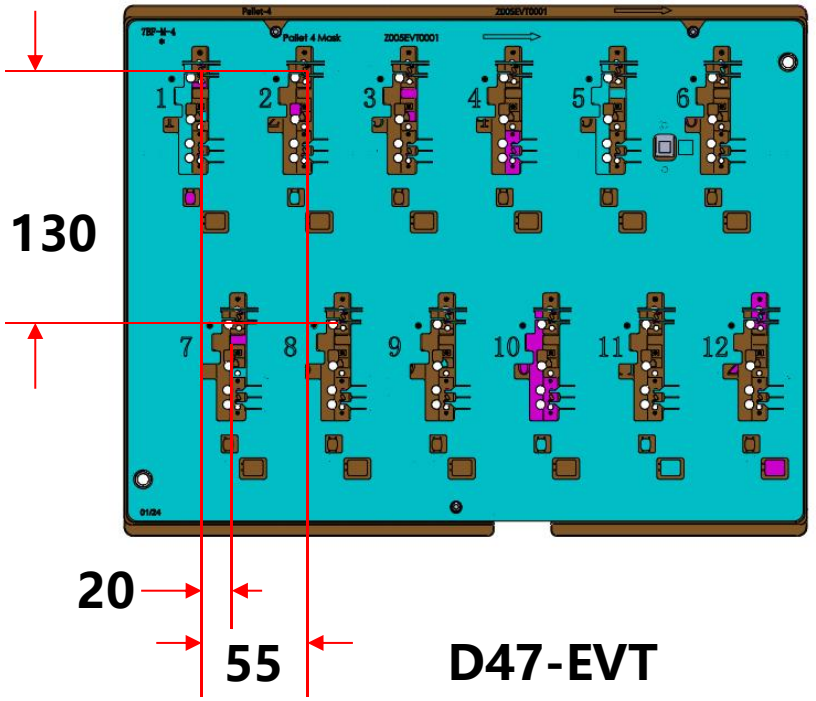
Spot 1&2 SUS(0.1) + SUS(0.28)

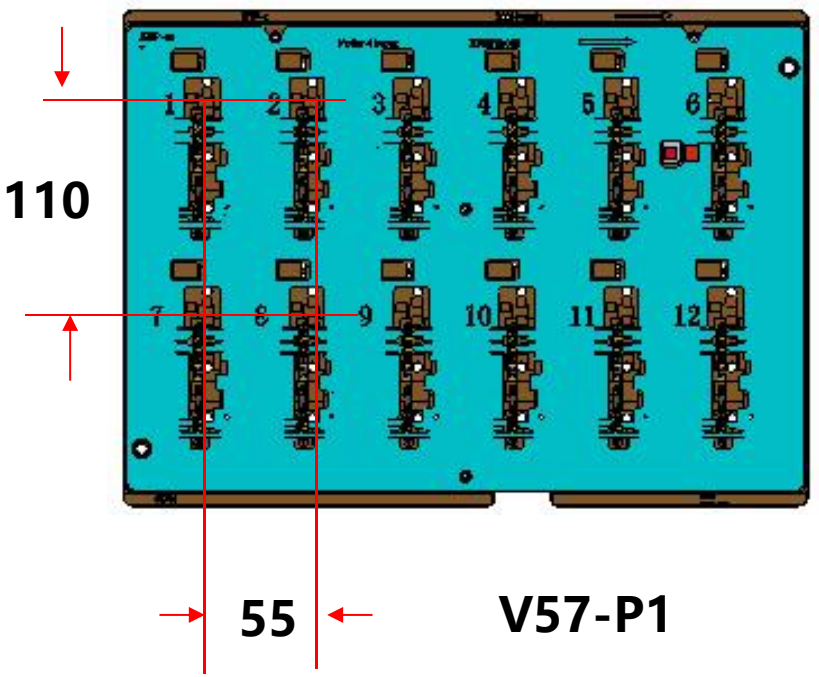
7.1 Welding Spot position



Item	Spot size(mm)	Distance in Y(mm)	Distance in X (mm)
S1	0.3	1.32	6.27
S2			
S3		1.7	
S4			

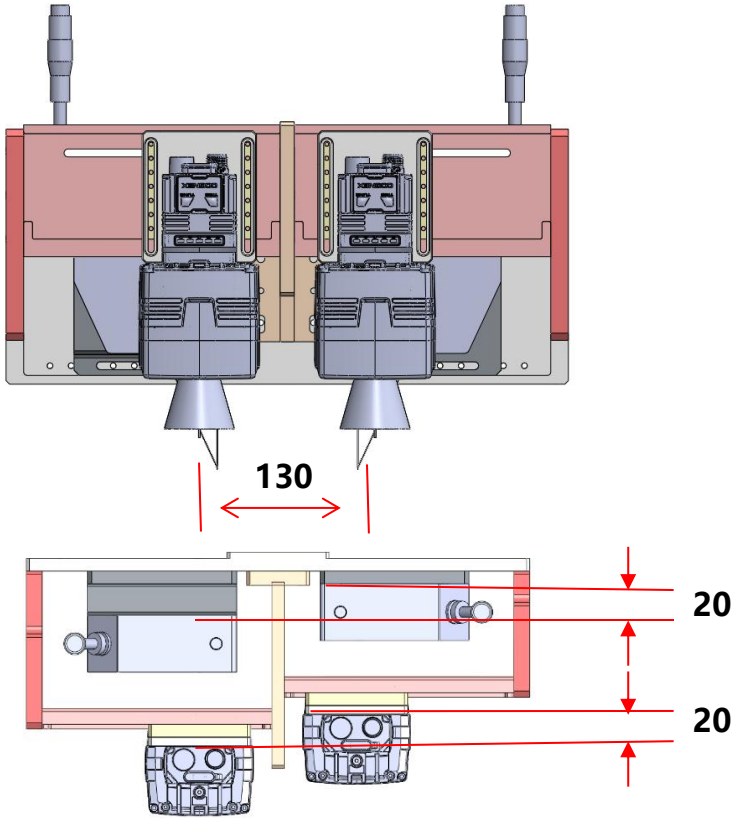
8 D47 EVT Carrier To V57 P1 Carrier Comparison



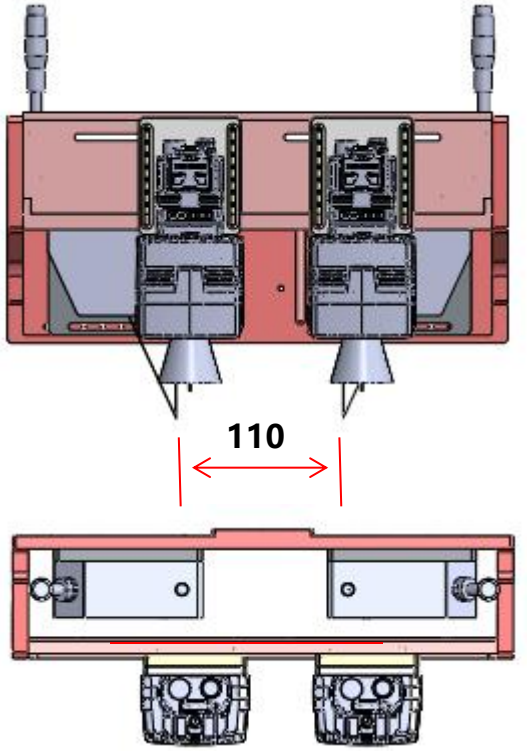


Item	overall dimension	Staggered L1 L2 L3 L4 in Y direction	Location holes position	Cavity numbers	Distance in Y direction (1~11) / (1~7)	Distance in X direction (1~2)
D47-EVT	360X280	Staggered 20 mm	340x220	2x6 total 12	130	55
V57-P1	360X260	Staggered 0 mm	340x200	2x6 total 12	110	55

9.1 Changes (scanner and height measure mounting bracket)



D47-EVT



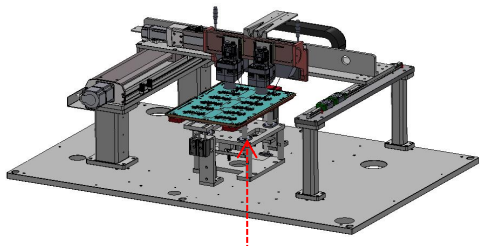
V57-P1

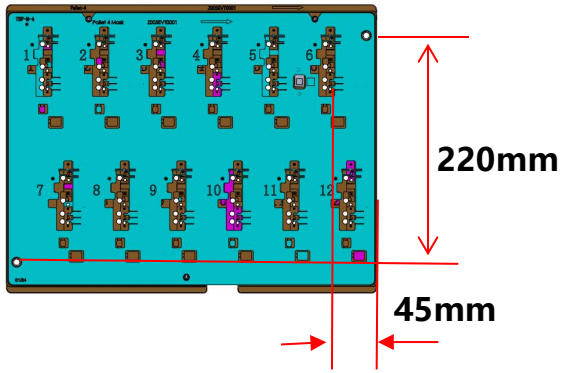
Action : The distance between the two Product is changed , the machine needs to be changed QR code scanner and height measure mounting bracket .

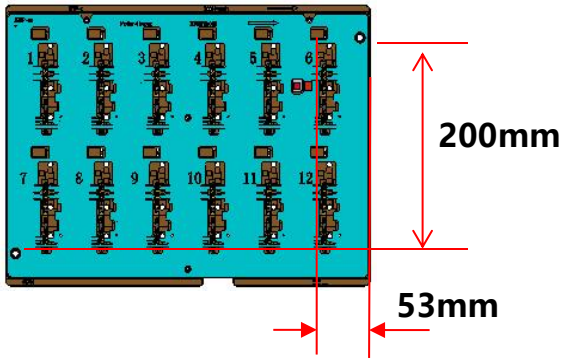
Hardware needed:

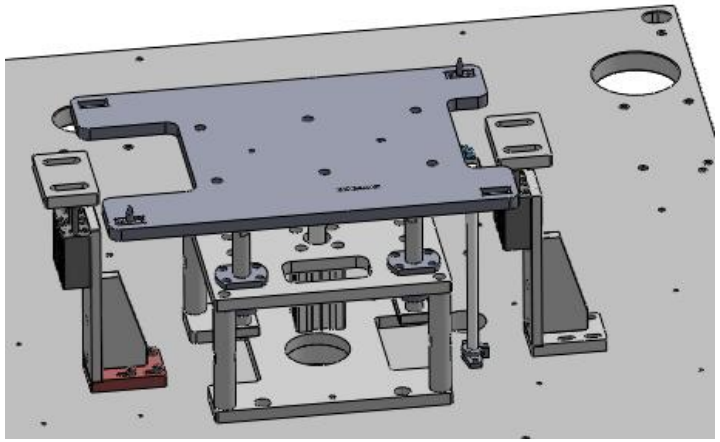
1.Scanning and height measuring module mounting kit*1

9.2 Changes(Product carrier mounting plate)



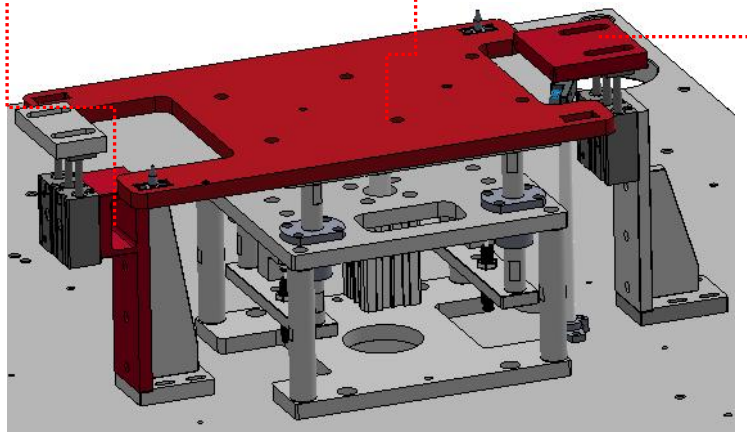






D47-EVT

Mounting plate | Support plate of the pallet | Stopper plate



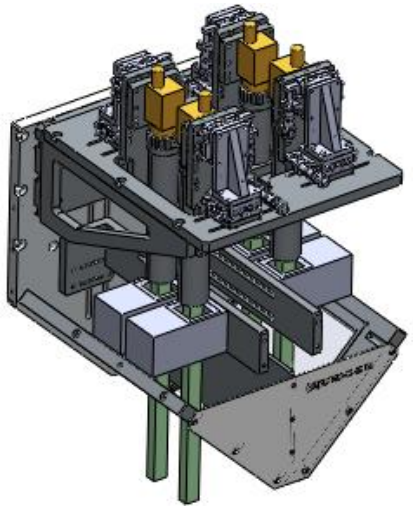
V57-P1

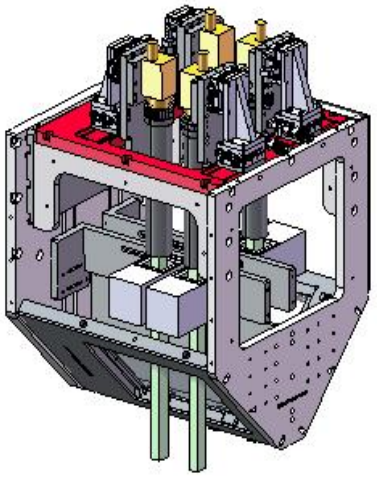
Action : the distance between the Location holes is from 220 to 200, and the distance between the last Barcode hole and the edge of the carrier is changed,the machine needs to be changed the mounting plate at the Barcode scan and height measure station .

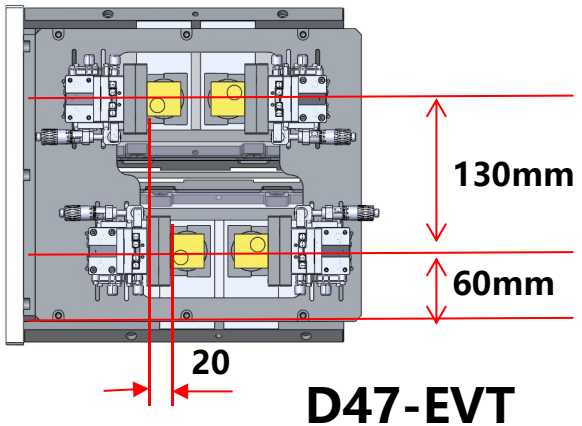
Hardware needed:

1.Product carrier module mounting kit*1.

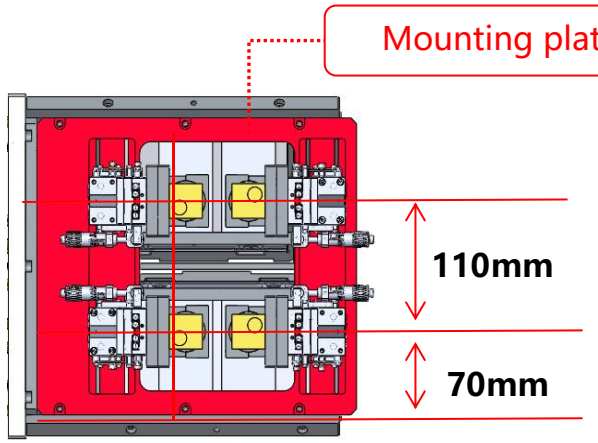
9.3 Changes (CCD mounting plate)







D47-EVT



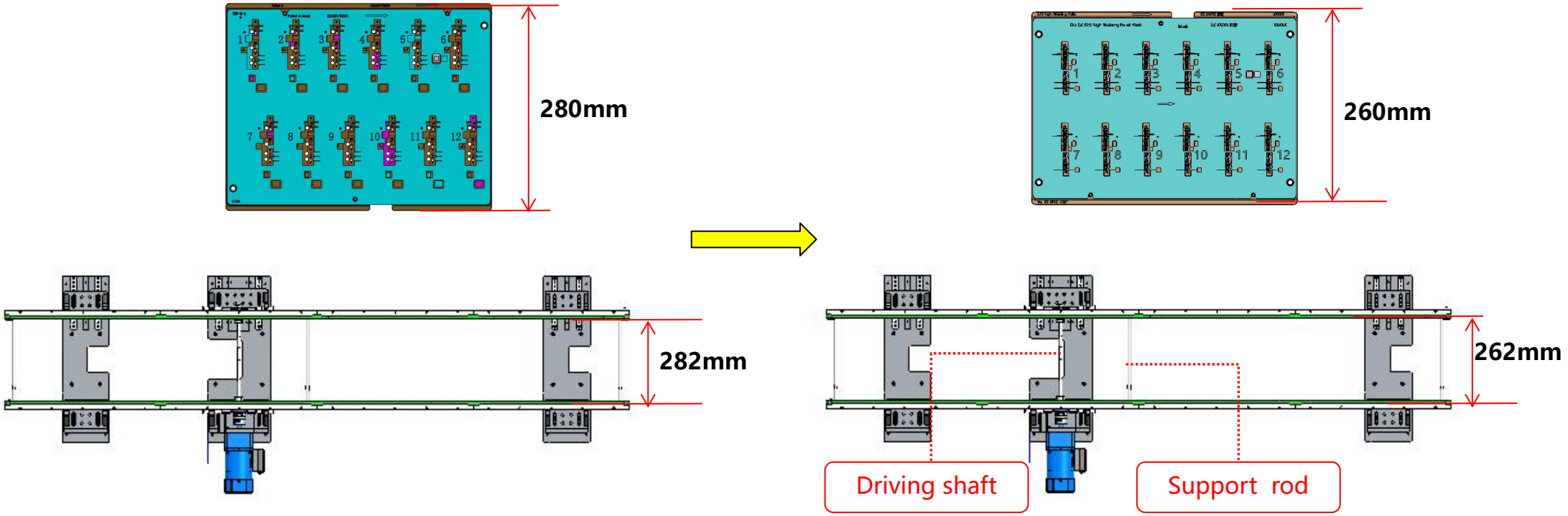
V57-P1

Action : The distance between the two Product is changed,the machine needs to be changed CCD mounting plate .

Hardware needed:

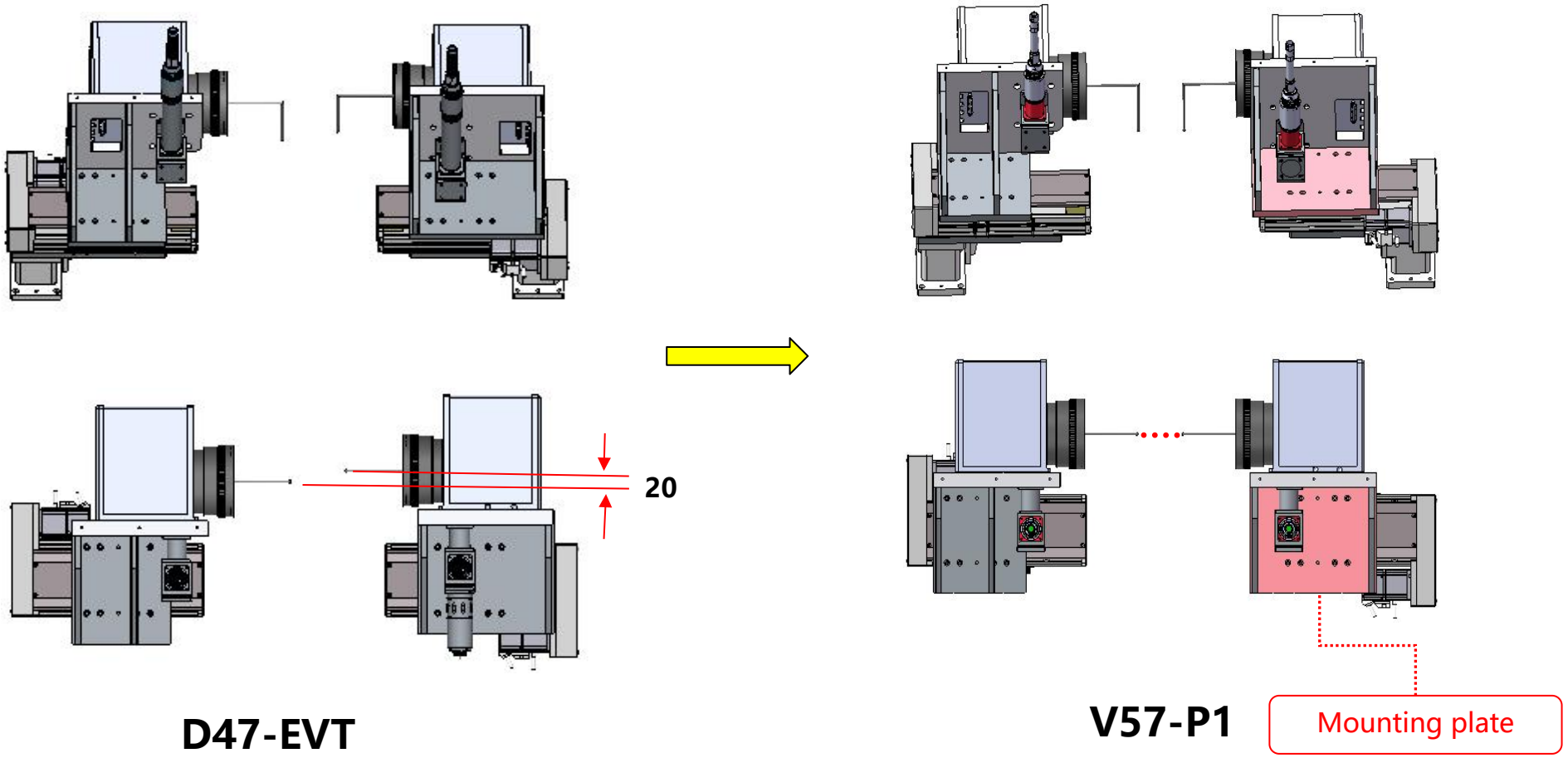
1.CCD module mounting kit*1.

9.4 Changes (the width of belt line)



Action : the size of product carrier is changed, the machine needs to be changed the width of belt line .
Hardware needed:
1.Support rod mounting kit*3. Driving shaft mounting kit*1

9.5 Changes （ Galvanometer Head mounting plate）

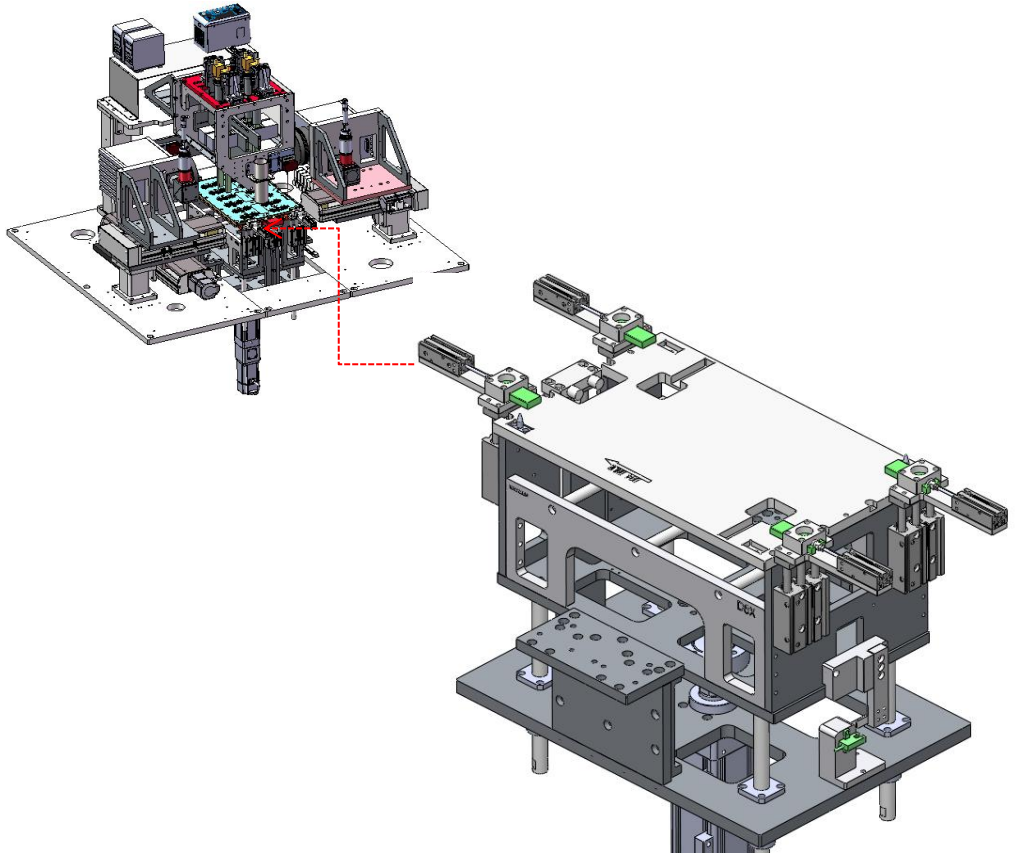


Action : the distance between the two Product is changed, the machine needs to be changed galvanometer head mounting plate .

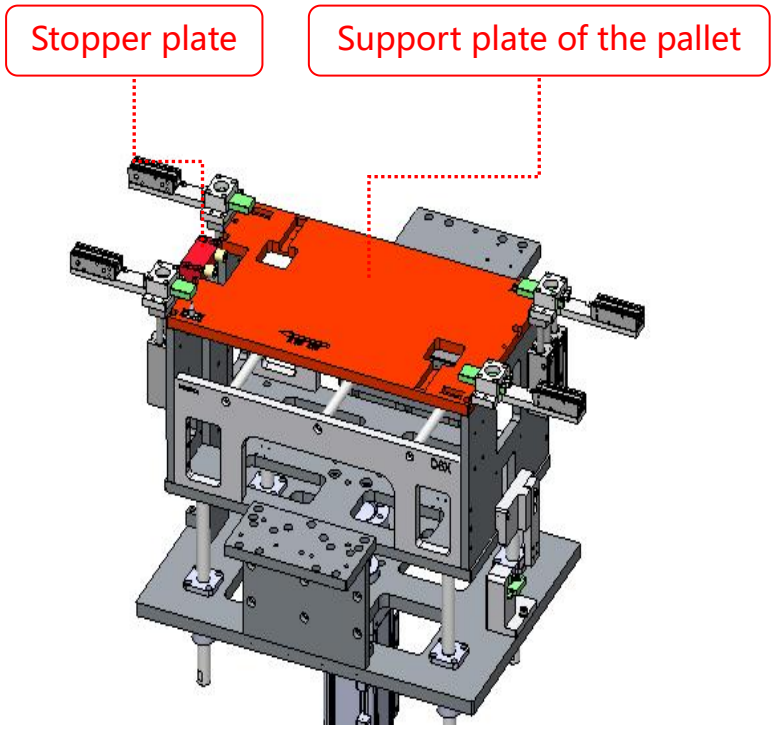
Hardware needed:

1.Galvanometer Head module mounting kit*1.

9.7 Changes (chang the plate)



D47-EVT



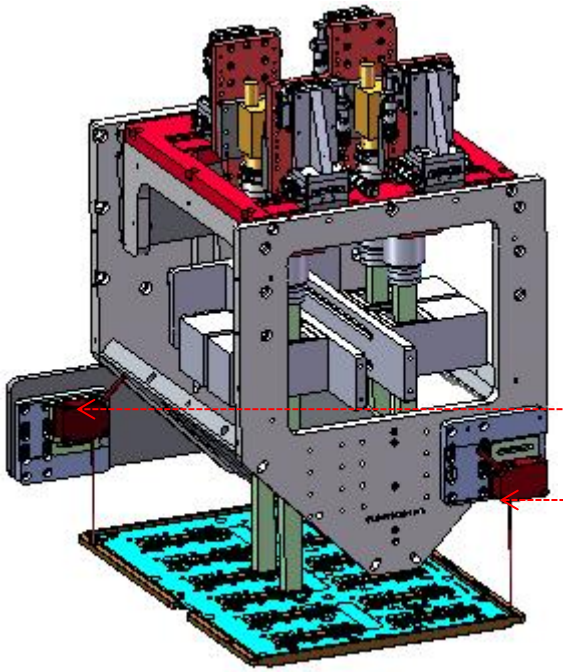
V57-P1

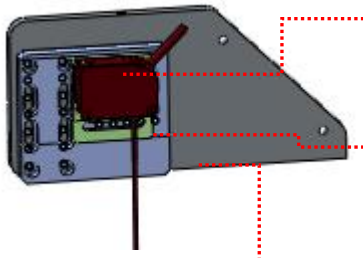
Action :the distance between the Location holes is from 220 to 200, the machine needs to be changed the mounting plate at the welding station .

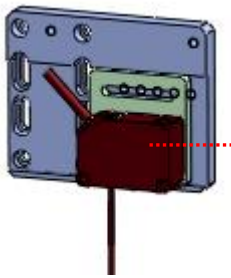
Hardware needed:

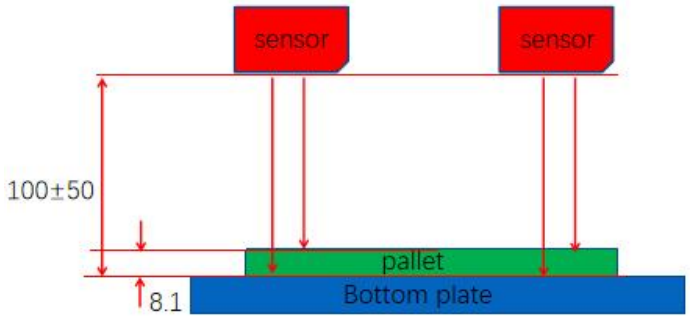
1.Product carrier module mounting kit*1.

9.8 Changes (add two laser sensors)



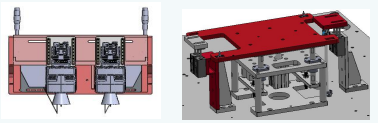
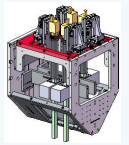
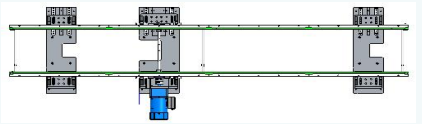
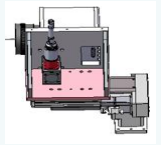
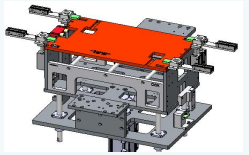




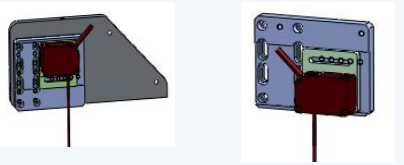


Action : add two laser sensors to check the pallet level.
Hardware needed:
1.Laser sensor *2 .
2.Laser sensor module mounting kit*2

Changes summary

Item	Description	Change reason	Image	Q'ty	Supplier
1	Barcode scan and height measure station —chang the mounting plate	the product carrier changed		1	HG
2	CCD station—chang the CCD mounting plate	the product carrier changed		1	HG
3	The width of belt line	the product carrier changed		1	HG
4	Welding station—chang the Galvanometer Head mounting plate	the product carrier changed		1	HG
5	Welding station—chang the mounting plate	the product carrier changed		1	HG

Changes summary

Item	Description	Change reason	Image	Q'ty	Supplier
6	Welding station—Distance measuring component	Increase stability		2	HG

Thank you for your attention!

HGTECH